Research Article

Preprint

# Not All Fakes Are Equally Ugly: The Anti-AI Bias in Aesthetic Judgment and the Role of Affect, Self-Relevance, and Authenticity

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osf.io/xbrhe

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Art is devalued once it is attributed to artificial intelligence (AI). Whether this anti-AI bias is a generic penalty for non-original works or specific to the absence of a human author is unknown, as is how long it lasts. In a preregistered study, 324 participants rated 48 genuine human paintings, each presented as an Original, a human Forgery or AI-Generated, on beauty, valence, meaningfulness and monetary worth, then judged how synthetic (AI vs. human) and authentic (copy vs. original) each was. Forgeries were rated below Originals on every dimension, and AI-Generated works about twice as low, the AI-specific penalty falling most heavily on meaningfulness. The labels did not alter gaze, and their effects did not depend on the emotional content or personal relevance of the artworks. Each label shifted the belief it implied, most clearly the AI label, and appraisal fed back into belief: beautiful artworks were judged human-made, even when never labelled. About six weeks later (n = 220), after learning that all artworks were human originals, participants no longer rated previously AI-labelled works as less beautiful. They had forgotten the labels but remembered what they had believed, and reconstructed the labels from these beliefs and their initial appraisal. The bias tracked attitudes toward AI, not interoceptive sensibility or visual imagery. The anti-AI bias thus combines a generic non-originality penalty with a penalty specific to the missing human author: it acts at the moment of judgment, yet leaves beliefs that outlast the memory of the label.

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: *Dominique Makowski*: conceptualization, methodology, software, supervision, project administration, data curation, formal analysis, investigation, visualization, writing – original draft, and writing – review & editing. *Ana Neves*: project administration and writing – review & editing. *Giulia Cabbai*: conceptualization and writing – review & editing. *Guido Corradi*: writing – review & editing. *Marco Viola*: writing – original draft and writing – review & editing. *Marco Sperduti*: project administration, writing – original draft, and writing – review & editing

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## Introduction

In October 2018, Christie's auctioned Portrait of Edmond de Belamy, an image produced by the French collective Obvious using a generative adversarial network, for over $400,000. Four years later, an image created with Midjourney by Jason Michael Allen won first prize in a digital art competition at the Colorado State Fair, triggering a public controversy about the legitimacy of algorithmically produced art. These two episodes crystallize a societal tension about the use of AI in the artistic domain: artificial intelligence (AI) is now able to produce images that enter markets, competitions, and museums on apparently equal footing with human productions, yet the same public that grants AI this economic and cultural legitimacy tends to devalue its output as soon as its algorithmic origin is revealed. A growing body of empirical work has documented this negative bias against art attributed to AI across a wide range of materials and paradigms. The effect has been reported for visual art (Bellaiche et al., 2023; Chiarella et al., 2022; Di Dio et al., 2023, 2026; Gangadharbatla, 2022; Ragot et al., 2020), for poetry and creative writing (Di Dio et al., 2026; Porter & Machery, 2024; Raj et al., 2023), for music (Ansani et al., 2025; Shank et al., 2022), and, with important cultural nuances, in cross-national comparisons of explicit and implicit attitudes (Wu et al., 2020). Raj et al. (2023) offer perhaps the most systematic demonstration of the phenomenon's robustness: across sixteen preregistered experiments and more than 27,000 participants, disclosing that a piece of creative writing had been produced by AI systematically lowered evaluations of its quality, creativity, and enjoyability, an "AI disclosure penalty" that resisted a wide array of moderation strategies, including collaborative human–AI framing and the humanization of the AI agent. A recent meta-analysis by De Rooij (2025), focused specifically on visual art, confirms that this devaluation is not an isolated laboratory curiosity: the effect is broad, generalizes across studies, and is moderated by factors such as participants' age, artistic style, the true source of the image, and the presentation context, but is consistently larger for judgments that require attributing meaning, intention, or depth to the artwork than for judgments of low-level sensory qualities.

This last observation already points toward the central puzzle that motivates the present work. If the bias were rooted in genuine perceptual differences between human and AI-generated images, it should scale with people's ability to actually tell the two apart. Empirically, the opposite pattern is observed. In blind evaluation conditions, without any information about authorship, observers are typically unable to discriminate AI-generated from human-made images above chance level (Gangadharbatla, 2022; Ragot et al., 2020). Zhou and Kawabata (2023) directly tested this dissociation using a free-viewing and categorization paradigm: participants were at chance level when categorizing AI-generated representational paintings as AI- or human-made, yet their gaze behavior differed between the paintings they categorized as human-made and those they categorized as AI-made, that is, according to believed rather than actual authorship. Porter and Machery (2024) report an analogous pattern for poetry: AI-generated poems were indistinguishable from human-written ones under blind evaluation, and in some conditions were even rated more favorably, before being devalued once their artificial origin was disclosed. The bias, in other words, does not track the observer's actual capacity to detect an algorithmic signature in the stimulus. It tracks belief.

This dissociation between perceptual indistinguishability and post-disclosure devaluation suggests that the anti-AI bias is better understood as a high-level, cognitive phenomenon than as a perceptual one. Direct evidence for this claim comes from studies that manipulate processing mode while holding the stimulus constant. Chiarella et al. (2022), testing visitors in an actual art exhibition, manipulated the attribution of authorship of two identical abstract paintings and found that the aesthetic devaluation of the "AI" painting emerged only in explicit comparative judgment; concurrent psychophysiological measures of affective engagement (electrodermal activity, heart rate) showed no corresponding modulation, suggesting that early, implicit affective responses to the image were left largely intact. Wang et al. (2026) manipulated authorship labels on images that were in fact all AI-generated and orthogonally manipulated whether participants evaluated them under time pressure and cognitive load (favoring automatic, stimulus-driven processing) or under conditions that encouraged reflective, meaning-oriented engagement. The AI label reduced appreciation in both conditions, but the effect was roughly four times larger under controlled, reflective processing. Similarly, work by Bara et al. (2025) on implicit and explicit measures of moral and aesthetic judgment toward AI art suggests that the bias operates mainly at an explicit, deliberate, context-sensitive level rather than as an automatic affective association. Converging evidence at the neural level comes from Zhang et al. (2025), who found that paintings labeled as human-made evoked greater activation of the right angular gyrus (a region implicated in semantic integration and episodic memory) than the same paintings labeled as AI-generated. Taken together, these findings converge on a single conclusion: the anti-AI bias is not primarily due to low-level perceptual stages of analysis, but a consequence of what the label "AI" does to the higher-order cognitive processes through which an observer constructs meaning, intention, and value around an aesthetic object.

This conclusion invites us to return to general cognitive models of aesthetic experience to ask where, exactly, in the chain of processing the label intervenes. In the influential model proposed by Leder et al. (2004) and updated by Leder and Nadal (2014), aesthetic experience unfolds through a sequence of stages: perceptual analysis, implicit memory integration, explicit classification, cognitive mastering, and a continuous affective evaluation process, all contributing to a final evaluation process resulting in two outputs, an aesthetic judgment and an aesthetic emotion. Two stages in this model are natural candidates for where an authorship label could exert its effect: explicit classification, where formal features of the artwork are related to prior knowl-

edge and expectations, and cognitive mastering, where the observer interprets the work in light of personal experience, biography, and expectations about the artist's intention. If information about AI authorship is integrated at these stages, it should selectively depress judgments that depend on inferred intentionality, meaning, and personal connection, while leaving more perceptually driven components of the aesthetic response comparatively untouched—precisely the pattern documented above. On the basis of this framework, and of the literature reviewed so far, at least three candidate mechanisms emerge as plausible drivers of the anti-AI bias: attributed originality and authenticity, the affective properties of the stimulus itself, and the personal relevance of the artwork to the observer. Each of these mechanisms motivates a specific methodological choice in the present study.

## Attributed originality and authenticity

The first candidate mechanism concerns the value historically attached to originals over copies. Newman and Bloom (2012), in an influential series of experiments, showed that people value original artworks over perceptually indistinguishable duplicates because of a principle of positive contagion: the original is believed to carry the physical and symbolic trace of the artist's body and intention, a trace that a reproduction cannot carry. A neuroimaging study by Huang et al. (2011) offers striking physiological support for this idea: when the same Rembrandt portraits were labeled as "copy" rather than "authentic", regardless of whether the picture was in fact genuine or false, viewing evoked stronger activation in frontopolar cortex and right precuneus, and the correlation between these regions and occipital visual areas increased. This was consistent with participants' reports of actively engaging effortful, hypothesis-testing processes to look for evidence supporting the assigned label. This shows that a mere verbal assignment of authenticity status is sufficient to reorganize the brain's engagement with an image, independently of its physical properties. At the same time, a meta-analysis by Specker et al. (2023) on the related "genuineness effect" - the presumed superiority of experiencing a physical original over a reproduction - found that the effect largely disappears once the confound between genuineness and museum-versus-laboratory context is properly controlled. The authors nevertheless caution that it is too early to conclude that the effect is absent, given the methodological limitations of the included studies, and suggest that a genuineness effect may emerge for more affective aspects of the aesthetic experience, which these studies have largely neglected. This body of work is important for the present study because it suggests that authenticity-related devaluation is not an all-or-nothing effect of "real versus fake", but is graded by what, specifically, the label communicates about the causal and intentional history of the object. Consistent with this graded view, Raj et al. (2023) show that the relation between AI disclosure and evaluative devaluation is itself mediated by shifts in perceived authenticity, further underscoring that the anti-AI bias and authenticity-related devaluation share a common underlying mechanism rather than being independent phenomena.

This distinction has a direct methodological consequence. Much of the AI-art literature has manipulated only a human-versus-AI authorship contrast (Bellaiche et al., 2023; Chamberlain et al., 2018; Gangadharbatla, 2022; Grassini & Koivisto, 2024; Ragot et al., 2020), which conflates two logically separable properties: not being an authentic original, and not being a human product at all. A painting attributed to a human copyist is also not the artist's original, yet it retains a human intentional history; a painting attributed to AI lacks both. Comparing only "human original" against "AI" therefore cannot determine whether devaluation reflects a generic non-originality penalty or a penalty specific to non-human authorship. To disentangle these components, the present study manipulates the presumed authorship of a set of artworks that were, in every case, actually created by human artists, assigning each piece one of three possible labels: human original, human copy, or AI-generated. This copy condition, directly inspired by the authentic-versus-copy manipulation of Newman and Bloom (2012) and Huang et al. (2011), makes it possible to isolate a bias that is specific to the AI label from a more general non-originality penalty, and thereby to test directly whether the anti-AI bias reduces to an authenticity problem or constitutes something more specific to algorithmic authorship.

Separating these two properties also raises the question of how each label is translated into beliefs about the artwork, and of what else informs these beliefs. If the forgery and AI labels communicate different things about an object's causal and intentional history, each should shift the belief dimension it semantically implies - authenticity (copy vs. original) for the forgery label, syntheticness (AI vs. human) for the AI label - rather than a single, undifferentiated sense of "fakeness". Beliefs about provenance, however, need not follow the label alone. Raj et al. (2023) showed that perceived authenticity mediates the effect of AI disclosure on evaluations, but the reverse route, from appraisal to belief, is also plausible: because observers cannot reliably tell AI-generated from human-made images by inspection (Gangadharbatla, 2022; Ragot et al., 2020), their judgments of origin must rest on indirect cues, and their own appraisal of the work is an obvious candidate. If AI images are expected to be less beautiful or less meaningful, a work that is experienced as such may be taken as more likely to be AI-generated or a copy. A label that depresses appraisal could then shift beliefs about origin not only directly, but also through the less favourable appraisal it induces, and appraisal could inform beliefs about origin even when no label is present. We therefore preregistered the prediction that more favourable appraisals (beauty, the extremity of valence, and meaningfulness) would predict judging an artwork as human-made and as original, and participants rated each artwork on the two dimensions of syntheticness and authenticity in a second phase of the experiment, after the aesthetic judgments. Decomposing the effects of the labels on these beliefs into direct and appraisal-mediated components goes beyond the preregistered analyses.

## Affective response in aesthetic evaluation: valence, arousal, and their interoceptive grounding

The second candidate mechanism concerns the affective response that an artwork elicits. Emotional processes occupy a central place in every stage-based model of aesthetic experience (Leder et al., 2004; Leder & Nadal, 2014), and the emotions elicited by an artwork are increasingly treated as integral to, and predictive of, its aesthetic appreciation (Menninghaus et al., 2019). Both the valence and the intensity of this response contribute. In our own group's related work, Salgues et al. (2024) showed that retrospective ratings of both the valence and the intensity of the emotional reaction to a painting tracked judgments of beauty, indicating that affective response is not merely a byproduct of aesthetic judgment but one of its structuring inputs. At the physiological level, Cabbai et al. (2023) found that skin conductance responses, a classical index of emotional arousal, directly and positively predicted both liking and being-moved ratings for visual artworks, independently of art type.

If affective response is one of the inputs of aesthetic judgment, an authorship label could lower appreciation by dampening the emotional response to the artwork. The evidence for such a dampening is, so far, limited and mixed. In a meta-analysis of the anti-AI bias in visual art, de Rooij (2025) found only small effects on outcomes reflecting the emotional and valuation components of aesthetic experience, compared with moderate effects on outcomes reflecting knowledge and meaning. Chiarella et al. (2022) recorded electrodermal activity and heart rate, following the classical proposal that arousal is better indexed by skin conductance and valence by cardiac measures, but, as noted above, neither was modulated by the authorship label, even though explicit aesthetic judgments were. Self-reported affect appears similarly resistant: Zhou and Kawabata (2023) found that subjective evaluations, including ratings of emotional valence and arousal, did not differ between the paintings that participants categorized as human- or AI-made, and Demmer et al. (2023) found that participants almost always reported emotions in response to abstract artworks, irrespective of whether they had been told that the works were made by a human or by a computer. Together, these findings suggest that the label leaves the early, bodily components of the affective response largely intact, but they leave open whether it alters how this response is appraised, the level at which aesthetic judgments are made.

A mechanism through which such a dampening could occur is suggested by research on fictional reappraisal, the implicit down-regulation of emotion that occurs when a stimulus is believed to be fictional rather than real. Emotional videos labelled as fictional elicited a weaker subjective emotional experience than the same videos labelled as real, although electrodermal arousal was unaffected (Sperduti et al., 2016), and negative pictures labelled as fictional were appraised as less intense and less negative, and elicited smaller skin conductance responses and late positive potentials (Makowski et al., 2019). An AI label conveys part of the same message: the image does not stem from a real person's experience and intention, and is in that sense not "real". If it is processed as a cue of unreality, it could trigger a similar down-regulation, attenuating the emotional response to AI-labelled works and, through it, their appreciation. Notably, this down-regulation was itself modulated by self-relevance: the personal memories evoked by a scene counteracted the effect of the fiction label (Sperduti et al., 2016), and the phenomenal and physiological changes depended in part on the self-relevance of the stimuli (Makowski et al., 2019), a link that anticipates the third candidate mechanism discussed below.

A growing body of work suggests that this affective dimension of aesthetic experience could be grounded, at least in part, in the perception of one's own bodily state - that is, in interoception. Cabbai et al. (2023) found that participants' interoceptive accuracy (assessed via a heartbeat-counting task) and interoceptive sensibility (assessed via self-report) selectively modulated emotional-intensity ratings for visual artworks, particularly in response to representational art. Using a very different kind of stimulus, Washizu et al. (2025) mapped the bodily sensations that accompany the aesthetic evaluation of everyday photographs and found that higher beauty ratings were specifically associated with chest-centered bodily sensations, that beauty tracked high-valence emotion categories, and that self-reported interoceptive awareness, while not directly related to beauty ratings, was nonetheless related to the overall intensity of reported bodily sensations. In the domain of immersive, participatory art, Kühnapfel et al. (2023) similarly found that visitors' bodily experience of an installation clustered into interpretable dimensions (including disturbance-related and interoceptive sensations such as heartbeat and breathing) that predicted artistic experience. The interoceptive cluster was specifically linked to interest. Together, these findings indicate that the affective response an artwork elicits is partly constituted by, or at least reliably co-occurs with, the observer's felt bodily state, and that stable individual differences in how people attend to and report on that bodily state can possibly modulate aesthetic experience.

Because affective response appears to be a driver of aesthetic appreciation, it is essential to control for it directly when comparing labeled conditions, otherwise, apparent "authorship effects" could in fact be confounded with pre-existing differences in the emotional profile of the stimuli themselves. For this reason, in the present study, we select stimuli drawn from a standardized set of paintings with normative subjective ratings, so as to sample systematically across the affective space defined by valence and arousal. This design choice allows us to enter normative valence and arousal ratings directly as predictors alongside the manipulated authorship label, and to ask whether affective properties modulate the anti-AI bias.

The evidence that emotional responses to art are partly embodied and interoceptively grounded also motivates a further, exploratory component of the present design. If, as Cabbai et al. (2023) and Washizu et al. (2025) suggest, individual differences in interoceptive sensibility shape how strongly bodily and affective signals feed into aesthetic eval-

uation, then such differences could plausibly also shape sensitivity to the anti-AI bias itself. On one hand, individuals who are more attuned to their own bodily and affective reactions might rely more heavily on this immediate, embodied signal when judging an artwork, and correspondingly less on the more abstract, belief-based inference triggered by an authorship label. This would predict a weaker label effect for participants high in interoceptive sensibility. On the other hand, insofar as interoceptive sensibility indexes a more general responsiveness to internally generated meaning, it could instead amplify the evaluative gap between conditions that succeed versus fail at engaging this internal resonance. Although the existing literature does not adjudicate between these possibilities, and no study to date has examined interoceptive sensibility in the specific context of authorship-based bias, we preregistered the second, namely that higher interoceptive sensibility would increase the difference between AI-labelled and other artworks. We include a brief, validated self-report measure of interoceptive sensibility to test it, and treat its relationship with aesthetic judgment more generally as exploratory. A second preregistered dispositional moderator concerned the vividness of visual imagery. Imagery and perception draw on overlapping sensory representations and are not always kept apart: people with more vivid imagery are generally more likely to report perceiving a stimulus (Dijkstra et al., 2021) and to mistake imagined content for reality (Dijkstra & Fleming, 2023), and imagery vividness is associated, albeit weakly, with phenomenological control, the capacity to generate experiences that match imaginative suggestions (Cabbai et al., 2024). If the authorship label acts as a prime that shapes how an artwork is experienced, its effect could be larger in people with more vivid imagery, and we preregistered that more vivid imagery would increase the difference between AI-labelled and other artworks.

## Self-relevance as a candidate mechanism of resonance

The third candidate mechanism is self-relevance: the degree to which an artwork connects to the observer's own autobiographical memories, identity, interests, and lived experience. The centrality of this dimension in aesthetic experience is well established independently of any question about AI. Vessel et al. (2012) earlier neuroimaging work had already shown that the most intensely moving aesthetic experiences recruit the brain's default mode network, a system associated with autobiographical memory and self-referential thoughts (Martinelli et al., 2013). Building on this tradition, Lee et al. (2023) exploited the self-reference effect in memory - the well-established finding that information encoded with reference to the self is subsequently better remembered than information encoded semantically or with reference to another person - as an implicit, objective index of aesthetic engagement: paintings that received extreme aesthetic judgments were remembered as well as paintings that had been explicitly encoded in a self-referential judgment task, suggesting that aesthetic judgment and self-referential processing draw on shared cognitive resources. Salgues et al. (2024) refined this result across two studies and showed that the mnemonic advantage conferred by aesthetic judgment is closely tied to both self-relevance and explicit emotional judgment, proposing that the perception of self-relevance may act as a switch that jointly triggers emotional engagement and memory integration, ultimately shaping the aesthetic judgment itself. Smith et al. (2025) provide complementary evidence that the spontaneous recollection of autobiographical memories during art viewing directly increases aesthetic appreciation, while Durkin et al. (2023), drawing on the art-historical concept of the "Beholder's Share", show that abstract artworks elicit greater interindividual variability in both verbal interpretation and default-mode network activity, but not in visual areas, than figurative artworks - evidence that the subjectivity of aesthetic experience is grounded in how observers mobilize internal, autobiographically structured representations rather than in low-level perceptual processes.

Most directly relevant to the present study is the work of Vessel et al. (2023), who showed that self-relevance predicts aesthetic appreciation not only for real artworks but also for synthetic images generated through neural style transfer, provided that these images are built around content that is autobiographically meaningful to the individual viewer: personalized synthetic artworks, constructed from photographs tied to each participant's own memories, places, and identity, were rated as aesthetically appealing as genuine human artworks, and considerably more so than synthetic images personalized for someone else. This result is important because it demonstrates that an algorithmically generated image can produce a full-quality aesthetic experience when its content, rather than its authorship, is calibrated to the observer's autobiography. It raises the natural question, central to the present study, of whether self-relevance can similarly attenuate or eliminate the anti-AI bias. This is why, in the follow-up session of our design, we directly measure participants' self-relevance judgments for each artwork, in order to test whether the label effect is moderated by, or interacts with, the degree to which an artwork resonates personally with the observer, and whether very high levels of self-relevance are associated with an attenuation of the gap between human- and AI-labeled artworks.

## Does the bias persist over time?

A question that has received comparatively little empirical attention concerns what happens to the anti-AI bias once the (misleading) information about authorship is no longer present, and whether any residual effect can be detected after a substantial delay. Several independent lines of reasoning motivate treating this as a central, if exploratory, research question in the present study. First, models of belief-based valuation such as Newman and Bloom's (2012) contagion account describe the value attached to an object as resulting from an inferred causal-intentional history rather than from its directly observable properties; there is no a priori guarantee that correcting a factual belief about that history

("this was actually made by a human") fully erases the representation that had been built around the original, false belief, particularly once the specific source of that belief has been forgotten. Second, the neuroimaging results of Huang et al. (2011) show that an authenticity label engages active, effortful hypothesis-testing and evidence-seeking processes rather than a passive updating of a simple prior; such elaborated, self-generated inferential processes are, in general, more resistant to correction than passively acquired beliefs, since forgetting the original source of a belief does not automatically remove the belief itself - a pattern well documented in the broader literature on source memory and belief perseverance (Johnson et al., 1993; Lewandowsky et al., 2012; Ross et al., 1975), and one that maps naturally onto our design, in which participants' explicit memory for the original labels can be assessed independently of their evaluative judgments. Finally, from an applied standpoint, understanding whether an initial "AI" framing produces a lasting evaluative trace, even once corrected, has direct implications for how disclosure and correction policies around AI-generated content should be designed and evaluated - an issue that is likely to grow only more pressing as generative systems become more capable and more widely used to produce cultural content.

For these reasons, our design incorporates a follow-up session in which participants who took part in the initial labeling study were invited back, after a substantial delay, to re-evaluate the same artworks - now correctly informed that all of them were, in fact, human productions. This follow-up session allows us to ask, exploratorily, two related questions: whether the evaluative difference induced by the original label persists once the correct information has been provided, and whether any such persistence is related to, or independent of, participants' explicit memory for the label that had originally been assigned to each artwork.

## The present study

Building on this theoretical and empirical background, the present study addresses five research questions. First, is the anti-AI bias reducible to a general authenticity or non-originality penalty, or does it reflect something specific to the attribution of algorithmic authorship, once a human-copy condition is included alongside human-original and AI conditions; and is each label encoded along the belief dimension it semantically implies, syntheticness for the AI label and authenticity for the forgery label? Second, to what extent do the affective properties of the artwork - its normative valence and arousal - account for aesthetic judgments independently of the attributed authorship label, and do they moderate its effect? Third, does the appraisal of an artwork inform beliefs about its origin, even in the absence of any label, and are the effects of the labels on these beliefs direct, or carried by the less favourable appraisal that the labels induce? Fourth, is the anti-AI bias moderated by the personal relevance of the artwork to the observer, such that high self-relevance narrows or eliminates the gap between human- and AI-labeled artworks? And fifth, does any label-induced bias in aesthetic judgment persist once participants are informed, after a delay, that all artworks were in fact of human origin, and how does this persistence relate to participants' explicit memory for the original labels and for their own beliefs? Finally, we examine how participants differ in their sensitivity to the labels, and test the preregistered predictions that positive attitudes toward AI would reduce, and higher interoceptive sensibility and more vivid visual imagery would increase, the difference between AI-labelled and other artworks. Addressing these questions jointly allows us to move beyond a simple documentation of the anti-AI bias toward an account of the specific cognitive mechanisms that structure it, and toward an initial empirical characterization of its persistence over time.

# Methods

## Participants

Participants were recruited via Prolific (n = 360). Thirty-six participants (11.1%) who failed at least half of the label catch trials were excluded, leaving a final sample of 324 participants (169 women, 153 men, 2 other; age M = 42.1, SD = 12.9, range 20–82), mostly residing in the United Kingdom (74%) and the United States (21%). Data from participants who failed the attention check embedded in a given questionnaire were discarded for that questionnaire only (MINT and VVIQ: n = 23; BAIT: n = 25). Syntheticness and authenticity ratings were additionally discarded for 7 participants who rated every image identically or whose response pattern was characteristic of a misunderstanding of the instructions. A subset of 220 participants completed the follow-up memory task, on average 47 days later (SD = 4.4, range 36–58). The study was approved by the University of Sussex ethics committee (ER/OO382/1).

## Procedure

The experiment was implemented with *jsPsych* (De Leeuw, 2015), and all materials, including its code, the items, instructions and exact procedure, are openly available at https://github.com/RealityBending/FakeArt, where the experiment can also be tested live.

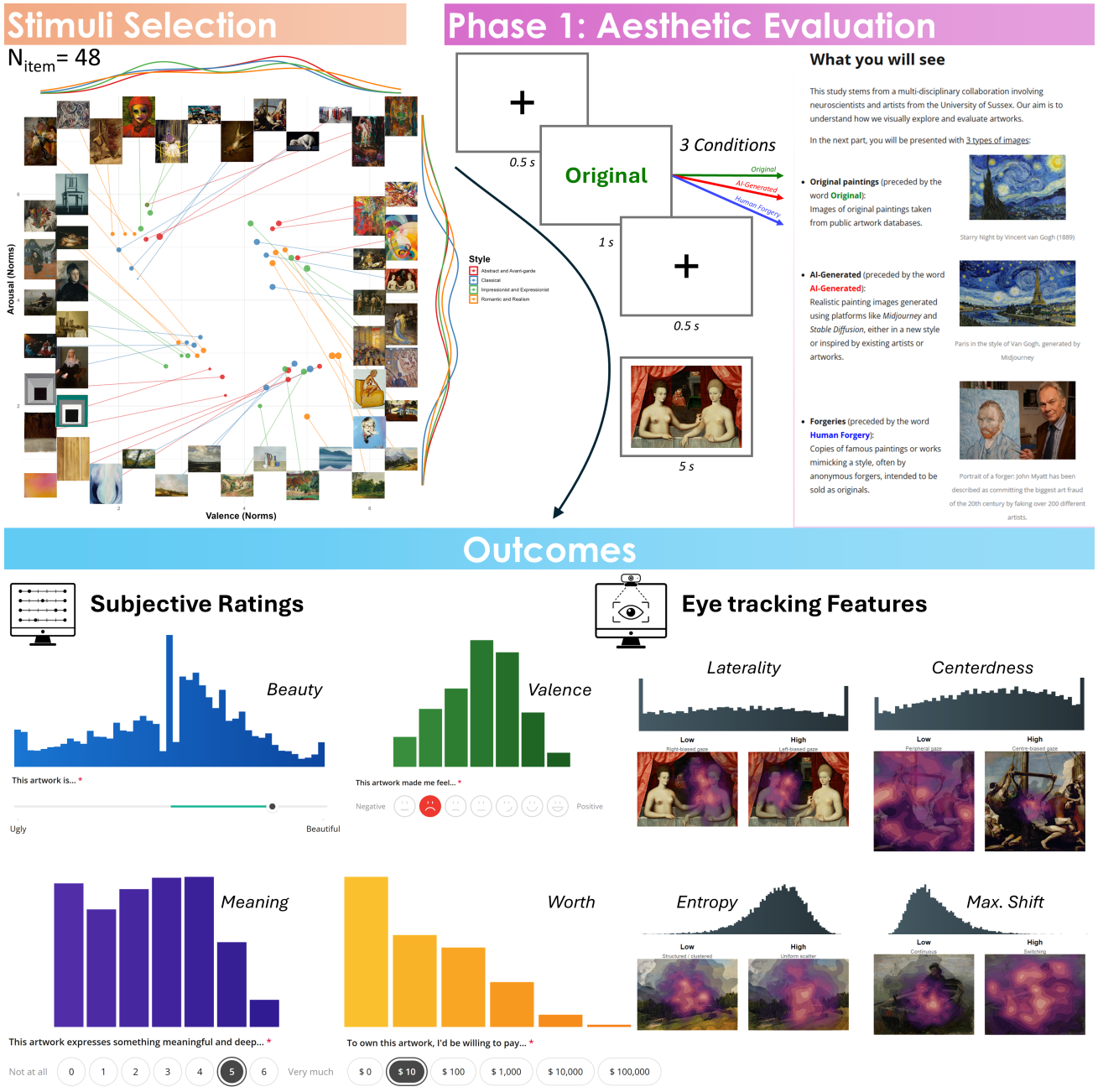
### *Aesthetic Judgment (Phase 1)*

Participants were first introduced to a cover story framing the experiment as a collaboration between neuroscientists and artists investigating how people visually explore and evaluate artworks (a framing that also justified the use of webcam-based eye-tracking, see below). They were told they would view three categories of images: Original Human paintings (sourced from public art databases), AI-Generated images (produced using generative platforms such as Midjourney and Stable Diffusion, either in a new style or inspired by existing artists or artworks[1]), and Human

[1]This intentionally equivocal description, that leaves open whether AI images were generated with the goal of being new or artist-inspired, was carefully considered and included to preempt the risk that a participant recognizing a genuine painting presented as AI-generated would undermine

**Figure 1**

*Overview of the study. Top left: selection of the 48 artworks from the Vienna Art Picture System by style (colours) and normative valence and arousal (axes), with the thumbnails of the selected paintings; the curves show the distribution of each style along the two norms. Top middle: a Phase 1 trial (fixation, 0.5 s; label, 1 s; fixation, 0.5 s; artwork, 5 s), followed by the four ratings; each artwork was preceded by one of the three labels. Top right: the description of the three categories of images given to participants in the cover story. Bottom: distributions of the four Phase 1 ratings with their response scales (left), and of the four webcam eye-tracking features (right), with example gaze heatmaps for low and high values of each feature.*




Forgeries (copies or stylistic imitations of existing artworks, historically produced by Humans with intent to deceive). Critically, only the participants' *beliefs* were manipulated - all stimuli were, in reality, genuine human-made artworks.

Stimuli from the Vienna Art Picture System (VAPS, Fekete et al., 2023) were first grouped into four broad stylistic categories (Abstract and Avant-garde; Impressionist and Expressionist; Romantic and Realism; and Classical). To minimize the influence of prior exposure, only paintings with low familiarity (familiarity ratings below the sample median) were retained. Within each style, the remaining paintings were stratified into four quadrants defined by the upper and lower tertiles of normative arousal and valence,

belief in the whole manipulation.

and the three paintings farthest from the style's median arousal/valence were selected from each quadrant. This procedure yielded a final set of 48 paintings - 12 per style, evenly split across the four arousal-valence quadrants (Positive - High Intensity; Positive - Low Intensity; Negative - High Intensity; Negative - Low Intensity). This *emotion category* was entered in the statistical models to test whether the label effects depended on the affective properties of the artworks.

During the first phase of the experiment, participants viewed the 48 artworks (see Figure 1), each preceded by a randomly presented label indicating its purported type ("Original", "Human Forgery" or "AI-Generated"). Following the presentation of each artwork (5 s), participants reported their aesthetic experience across four dimensions encompassing cognitive appraisal, emotional response, semantic processing, and valuation (Kirk et al., 2009; Leder et al., 2004; Leder & Nadal, 2014): *beauty* (aesthetic quality, composition, and execution), emotional *valence* (the intensity of positive or negative feelings evoked), *meaningfulness* (conceptual richness, symbolism, and profundity), and monetary *worth* (maximum willingness to pay). To mitigate common method bias (Podsakoff et al., 2003) and minimize construct overlap, each dimension was assessed with a bespoke scale: an analog slider (Ugly - Beautiful), a 7-point pictorial scale (Negative - Positive), a 7-point numerical scale from 0 (Not at all) to 6 (Very much), and a 6-point discrete logarithmic monetary scale offering choices from \$0 to \$100,000[2]. To ensure sustained engagement with the contextual cues, random catch trials were interleaved throughout the block, prompting participants to recall the immediately preceding label.

At the end of Phase 1, participants completed a brief manipulation check questionnaire, framed as a feedback opportunity. Checklist items probed their overall impressions of the relative aesthetic quality of AI-generated versus human-made images, the perceived convincingness of the forged artworks, and the degree to which they trusted the labels (e.g., whether labels seemed mismatched, reversed, or whether they suspected that all images in fact belonged to a single category). Participants who selected believing that every artwork was either genuine or AI-generated were additionally asked to rate their confidence in that belief, which was used as a manipulation distrust proxy. This questionnaire was implemented to help identify suspicious or disengaged participants.

## *Questionnaires*

Between the two phases of the artwork judgment task, participants completed four questionnaires. The *Beliefs about Artificial Images Technology* scale (BAIT) was originally developed in Makowski, Te, Neves, Kirk, et al. (2025) to assess expectations regarding the realism of computer-generated content and iteratively extended across subsequent studies. The version administered here (BAIT-14) comprised 14 items spanning two components: beliefs about the realism of AI-generated images, videos, text, and artworks (e.g., "Current AI algorithms can generate very realistic images", "AI-generated art can sometimes surpass human creativity and artistic value"); and general attitudes toward AI, adapted from four items of the General Attitudes towards Artificial Intelligence Scale (GAAIS, Schepman & Rodway, 2020) capturing perceived danger, worry, excitement, and societal benefit. The questionnaire additionally included two items assessing self-rated AI knowledge and frequency of AI tool use, and one attention check item.

The *Multimodal Interoception Questionnaire* [MINT; Makowski, Neves, Benn, et al. (2025)] is a 33-item self-report measure of interoceptive sensibility spanning eleven bodily domains, including cardiac, respiratory, gastric, dermal, urogenital, sexual, and olfactory sensations, as well as satiety and excretory awareness. Each domain was scored as the average of three items (e.g., "I can notice even very subtle changes in the way my heart beats"). One additional item served as an attention check.

Participants also completed the *Vividness of Visual Imagery Questionnaire* (VVIQ, Marks, 1973), a 16-item measure of the subjective vividness of voluntary visual imagery. Participants were asked to visualize four scenarios in turn (a familiar person, a rising sun, a familiar shop front, and a countryside scene) and, for each of four component images within a scenario, rated the vividness of the resulting mental image.

Finally, participants completed the four-item *Patient Health Questionnaire* (PHQ-4, Kroenke et al., 2009; refined version: Makowski, Te, Neves, & Chen, 2025), a brief screening measure of anxiety and depressive symptoms. Participants additionally rated their global life satisfaction on a single item ("All things considered, how satisfied are you with your life as a whole?" Cheung & Lucas, 2014).

## *Reality Beliefs (Phase 2)*

At the start of the second phase, the cover story was *partially* revealed[3]. Participants were informed that some of the Phase 1 labels had been intentionally randomized and were tasked with identifying the true nature of the artworks based on their own judgment (preserving the belief that originals, forgeries and AI-generated stimuli were present). For each briefly re-presented image (1.5 s), participants made two continuous, gut-feeling judgments, probing theoretically distinct aspects of reality beliefs: *syntheticness* ("AI-generated" vs. "Human Creation") and *authenticity* ("Copy/forgery" vs. "Original Creation"). We treated these two dimensions as orthogonal, such that artworks could in principle occupy any of four categories (Human Original, Human Forgery, AI Original, AI Copy) - the same four-category structure later probed directly in the follow-up memory task[4]. Finally, participants were debriefed about

[2]The selection, description, wording and presentation of these measures were refined through extensive pilot testing and expert feedback.

[3]This pseudo-reveal paradigm was developed as an improvement over other designs in related experiments, see **FICTIONERO PREPRINT REF** for discussion

[4]Ideally, this orthogonal structure would have been also reflected in Phase 1 through four corresponding labels rather than three. We collapsed

the true nature of the experiment and the fact that all stimuli were Human artworks.

### *Follow-up: Self-relevance and Memory Task*

Participants were invited for a follow-up session approximately six weeks after their initial participation (see Participants for the exact delays). They were first reminded of the two-phase label-manipulation procedure from the original session and we explicitly re-iterated that all previously viewed artworks had, in reality, been genuine human-created originals. They then completed a recognition-memory task in which the original 48 artworks were randomly intermixed with an equal number of novel, previously unseen artworks, presented in randomized order. Novel artworks were drawn from the same VAPS stylistic pool and matched one-to-one to old artworks within each style by minimizing the total Euclidean distance across standardized ratings of valence, arousal, liking, complexity, and familiarity; Bayesian *t*-tests confirmed that old and new items did not differ on any of these norms (all BFs < 1).

For each artwork, participants first rated its beauty once again (using the same slider as in Phase 1) and its personal/self-relevance (the extent to which the artwork related to their own experiences, personality, or life events, independently of how beautiful they found it). They then made an old/new recognition judgement indicating whether they recalled the artwork from the original session. Artworks judged as novel were additionally rated for perceived artificiality (i.e., how easily participants could believe the image was AI-generated). Artworks judged as previously seen instead prompted two source-memory questions: which condition label (Original, AI-Generated, or Human Forgery) was associated with the artwork in Phase 1, and what was their own reality belief response made in Phase 2. Namely, participants had to select which of four categories participants had endorsed, corresponding to the combination of the dichotomized syntheticness and authenticity ratings (Human Original, Human Forgery, AI Original, AI Copy). This design jointly captured item memory (recognition) and source memory in its external and contextual dimension (recall of the assigned label) and internal aspect (one's own past belief).

## Data Analysis

### *Eye Tracking*

Gaze position was sampled continuously via webcam-based eye-tracking (WebGazer.js) throughout each trial and re-expressed in coordinates centered on the stimulus and normalized by stimulus width, preserving the image's aspect ratio and allowing comparison across images of different sizes. Of the 324 participants of the final sample, 318 provided usable eye-tracking recordings. Given the noisy and variable nature of webcam-based data, a strict quality control workflow was applied. 15 participants (4.7%) were excluded due to invalid data (>25% consecutive duplicate samples). Off-screen coordinates and samples reflecting implausibly fast transitions (>6 stimulus-widths/second, indicative of blinks or artifacts) were removed. Finally, trials were excluded if they contained fewer than 20 valid samples, showed frozen (SD < 0.02) or excessively noisy (SD > 0.60) gaze on either axis, had less than 50% of samples falling within the stimulus bounds, exhibited severe linear drift (>1 stimulus-width of total displacement), or showed near-uniform spatial scatter consistent with untracked gaze. Finally, 18 participants were excluded for yielding fewer than 2,000 valid samples across the full experiment. Together with 11 participants whose trials were all removed by the trial-level criteria, this left 12,220 trials from 274 participants (84.6% of the final sample) for the gaze analyses.

Four simple, robust and interpretable gaze features were computed per trial. *Laterality* was the proportion of samples falling in the left half of the image (vs. the right), and *Centeredness* was the proportion of samples falling within the center portion of the image. *Entropy* quantified the spatial dispersion of gaze: samples were binned into a 5x5 grid, Shannon entropy was computed and normalized by the maximum entropy achievable given the number of valid samples (avoiding artificial deflation on sparse trials). *Max. Shift* captured the single largest reorganization of gaze focus within a trial: for every possible partition of the chronologically ordered samples into an initial and final segment (each with at least 30 samples), the Euclidean distance between the two segments' centroids was computed, and the maximum across all partitions was retained; large values indicate a relocation of attention to a different image region, and near-zero values indicate a single stable focus throughout.

As a sensitivity check, we verified that these features - derived from noisy, low-cost webcam data - nonetheless carried signal relevant to the behavioural task, by regressing each of the eight rating outcomes (the four Phase-1 aesthetic dimensions, the two Phase-2 belief judgments, and the beauty and self-relevance ratings of the follow-up) on each of the four gaze features in turn (mixed models with random intercepts for participant and item, controlling for the number of valid gaze samples of the trial). All four features showed at least one significant association, expressed in percent of the outcome's scale range per SD of the gaze feature. *Entropy* was the most broadly sensitive feature, positively predicting meaningfulness (0.73% [0.31, 1.15]), authenticity (0.63% [0.08, 1.18]), self-relevance (0.60% [0.12, 1.09]), beauty (0.57% [0.23, 0.92]), worth (0.42% [0.07, 0.76]) and valence (0.38% [0.01, 0.74]), indicating that a more widely distributed exploration of an artwork accompanied more positive appraisals. Conversely, *Max. Shift* was negatively associated with authenticity (-0.66% [-1.22, -0.10]), valence (-0.41% [-0.78, -0.03]) and worth (-0.40% [-0.75, -0.04]), suggesting that an abrupt relocation of gaze partway through a trial accompanied less favourable eval-

this to three categories for practical reasons: presenting an artwork under a label such as "AI-generated original" risked a participant recognizing it as a known human painting, which would undermine trust in the labelling manipulation as a whole. Piloting also suggested that the three-category framing was more intuitive and more ecologically valid, as AI-generated and forgeries typically function as stand-alone categories in everyday life

uations. *Laterality* predicted self-relevance (1.11% [0.54, 1.68]) and beauty at follow-up (0.62% [0.11, 1.13]), and *Centeredness* predicted syntheticness judgments (-0.66% [-1.26, -0.07]). All these associations are very small - at most about 1% of the scale range per SD - but still are evidence that the gaze features are not pure noise.

### *Statistical Models*

Given the varied nature of the outcome scales, model families were chosen to match each outcome's distributional characteristics rather than defaulting to linear (Gaussian) models, which assume continuous, unbounded, unimodal outcomes and would poorly capture the shape of most of our data. *Beauty*, *Syntheticness* and *Authenticity* all displayed the characteristic bimodal pattern with inflation at 0, 1 and the midpoint, consistent with a Choice-Confidence process in which a discrete evaluative choice (e.g., "Ugly vs. Beautiful", "Human vs. AI") is conflated with a continuous rating of degree. These outcomes were modelled with a two-component Beta mixture that explicitly parameterizes this two-stage process [the Choice-Confidence model, CHOCO; Makowski, Neves, and Field (2025)], implemented in the *cogmod* R package (Makowski, 2025). *Valence* ratings were bounded and discrete, and thus modelled with a Beta-Discrete distribution, which accommodates the discretized nature of the response scale (Sciandra et al., 2024). *Meaning* showed additional mass at zero beyond what the Beta-Discrete component could capture and was modelled with a zero-inflated extension. *Worth* and *Self-Relevance*, whose response options form discrete, ordered categories with idiosyncratic boundaries between adjacent levels (e.g., the distinction between $1,000 and $10,000, or between a Self-Relevance rating of 1 and 2 can be variable between participants), were modelled with Cumulative (ordinal) models. Among the eye-tracking features, *Entropy*, a continuous bounded proportion, was modelled with a Beta distribution, while *Laterality* and *Centeredness*, which additionally showed boundary inflation at 0 and 1, were modelled with Zero-One-Inflated Beta (ZOIB) distributions. *Max. Shift*, strictly positive and right-skewed, was modelled with a LogNormal distribution.

Using bespoke models suited to each outcome's characteristics respects features of the data that linear models ignore (bounded scales, zero-inflation, bimodality, ordinal categories), and makes it possible to separate distinct components of the responses. For instance, CHOCO models allow to disentangle effects on the probability of each discrete choice, from effects influencing the degree (mean and spread) of each choice separately. For simplicity, we will however focus on reporting effects on the main parameters (typically $\mu$), reflecting shifts in the outcome's central tendency. For models with additional parameters of theoretical interest (e.g., *p*-Zero in the zero-inflated Beta-Discrete model for Meaning), effects on these parameters are also reported and interpreted where relevant. Effects are reported as marginal contrasts between label conditions on the response scale (posterior median with its 95% HDI as CI), averaged across emotion categories and, to assess the consistency of the effects, within each emotion category, and an effect is considered significant when its CI excludes zero. To make effect sizes comparable across outcomes, all differences are expressed in percent of the scale's range, or choice probability depending on the parameter. In the main models, the outcome (and each modelled distributional parameter) was predicted by the label condition (with Human Originals as the reference level), the emotion category of the artwork, and their interaction, with random slopes for the label condition, the emotion category and their interaction over participants, and random slopes for the label condition over artworks (the emotion category being a property of the artwork). Gaze models additionally controlled for the number of valid samples of each trial.

To test whether the label effects on reality beliefs were carried by the Phase 1 ratings, syntheticness and authenticity were also modelled as a function of the label, the Phase 1 beauty rating of the same trial (centered within participant), and their interaction. Because the label was randomized and beauty was rated before the beliefs, each label effect could be decomposed into a *direct* effect and an *indirect* effect carried by the change in beauty that the label induced (Imai et al., 2010; Pearl, 2001), computed from population-level predictions at the average beauty under each label. The same decomposition was repeated with the four Phase 1 ratings as joint mediators, yielding the *unique* indirect effect of each rating, and with the follow-up beauty rating (unaffected by the labels) as a covariate, in the participants who completed the follow-up. Finally, we modelled the perceived artificiality of the artworks judged as new at follow-up as a function of their follow-up beauty, and reality beliefs as a function of the style and normative ratings of the artworks (Fekete et al., 2023). As exploratory checks of the model specification showed a non-linear (convex) relation between beauty and authenticity, the authenticity model was also fitted with a quadratic term for beauty.

Self-relevance, rated after the debrief and unaffected by the labels (see Results), was treated as a label-free moderator rather than as a mediator, and centered within participant, so that its effects contrast an artwork with the participant's other artworks. Phase 1 beauty and meaningfulness were modelled as a function of the label, self-relevance and their interaction, comparing the label contrasts between artworks one SD less and more self-relevant than the participant's average; the moderation of beauty was also tested with follow-up beauty as a competing moderator, and with the participant's mean self-relevance as a between-participant moderator. Self-relevance was also added, alongside Phase 1 and follow-up beauty, to the models of reality beliefs and of perceived artificiality described above.

To look for a residual label effect at follow-up, follow-up beauty was modelled as a function of the label and of whether the artwork was recognized, and as a function of the label and of the Phase 2 syntheticness belief (controlling for Phase 1 beauty), the latter decomposing the label effect into the part carried by the belief and the rest, as above. Memory was modelled with categorical (multinomial logit)

models of the answer given to each artwork, in which "not recognized" (the artwork judged as new) was one of the possible answers alongside the three labels (or the four belief categories), so that recognition and source memory were estimated jointly, with participant- and artwork-level random effects. Because the new artworks form an additional level of the label predictor, the answer probabilities of these models recover the quantities of a signal-detection analysis, namely hit and false-alarm rates, source accuracy and response biases towards particular answers, as functions of the predictors and without collapsing the answers into a binary decision. The recalled label was predicted by the label shown (new artworks forming an additional level, which captures false alarms), the recalled belief by the participant's actual Phase 2 belief (dichotomized on both sliders), and the recalled label by both the label shown and the belief, to test whether labels were reconstructed from beliefs. A fourth model predicted the answers given to previously seen artworks from the label and from the Phase 1 beauty and valence ratings, each centered within participant and entered with a quadratic term, to test whether extreme appraisals were better remembered. To test whether the correspondence between recalled and actual beliefs reflected memory rather than a fresh inference from the current impression of the artwork, the models of the recalled belief and of the recalled label were also fitted with the Phase 1 beauty, follow-up beauty and self-relevance of the artwork as covariates (centered within participant). The recalled label was further modelled on all artworks as a function of self-relevance and follow-up beauty (with quadratic terms), with separate slopes for old and new artworks, so that an effect on hits could be told apart from an effect on false alarms. In the memory models, effects are differences in the probability of each answer.

Interindividual differences were examined on participant-level indices derived from the random effects of the main models. For each participant and distributional parameter, we computed, on the link scale and averaged over the emotion categories, the value of the parameter for Originals (*baseline*) and its difference between Forgeries or AI-Generated images and Originals (*label effects*). We also extracted each participant's slope of syntheticness on Phase 1 beauty (the "beautiful = human" inference described in the Results) and, from the categorical model of the recalled label, their tendency to attribute each label to the artworks they recognized. The reliability of each index was estimated with D-vour (Lüdecke et al., 2021), which corresponds to the variance of the participants' posterior means relative to the sum of this variance and the squared average posterior SD, with values above .75 indicating strong, around .5 moderate, and below .5 weakly identified individual differences. The structure of the Phase 1 and Phase 2 indices with a reliability above 1/3 was examined with a hierarchical Exploratory Graph Analysis (EGA, Golino & Epskamp, 2017; Jiménez et al., 2023), which estimates a regularized partial-correlation network of the indices, partitions it into lower-order communities (Louvain algorithm with consensus clustering, Blondel et al., 2008), and groups these into higher-order communities (Leiden algorithm, Traag et al., 2019); its stability was assessed with bootstrapping (Christensen & Golino, 2021). The network scores of the lower- and higher-order communities and the label recall tendencies were then correlated with the participants' demographic and questionnaire characteristics (Pearson correlations), with a false-discovery-rate (FDR) correction across all the correlations reported (Benjamini & Hochberg, 1995).

In the Results, numerical estimates are given for the main effects only; the estimates of all other effects mentioned, and all analyses, are openly available at https://github.com/RealityBending/FakeArt.

# Results

## Effect of Label Condition on Aesthetic Judgments

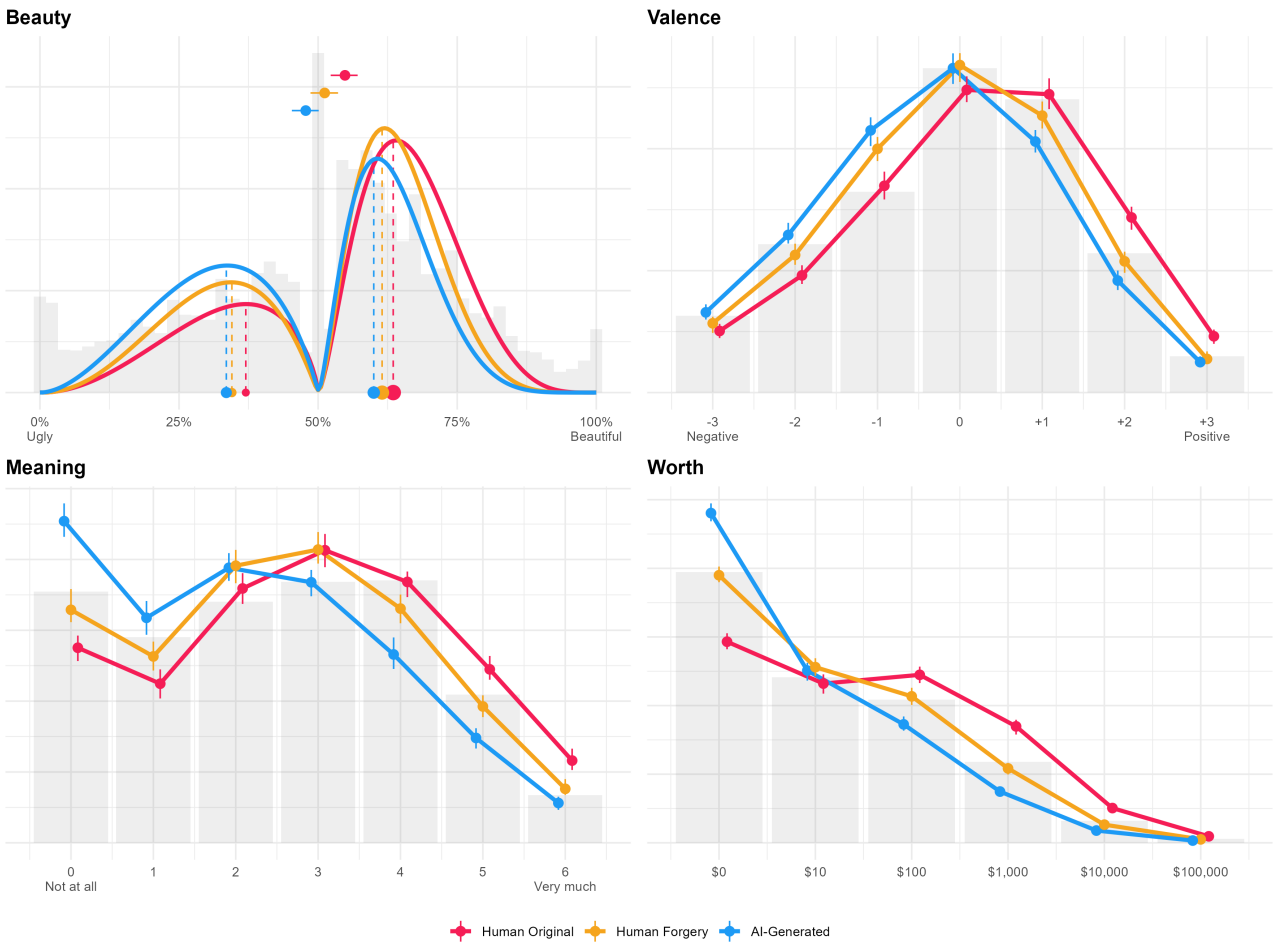
Across the Phase 1 ratings, the three labels ordered consistently: artworks presented as Human Forgeries were rated in-between Human Originals and AI-Generated images, with AI-Generated images receiving the lowest ratings on every dimension (see Figure 2).

Compared to Human Originals, artworks labelled as Forgeries were judged less beautiful (-3.7% [-4.6, -2.8]), as evoking less positive emotions (-3.5% [-4.3, -2.7]), as less meaningful (-4.3% [-5.3, -3.3]) and as less valuable, with a higher probability of being assigned no monetary value at all (+9.5% for the "$0" option [8.3, 10.6]). The AI-Generated label produced the same pattern with roughly twice the magnitude (Beauty: -7.1% [-8.2, -5.9]; Valence: -6.2% [-7.2, -5.1]; Meaning: -10.4% [-11.9, -9.0]; "$0" option: +18.3% [17.1, 19.5]), and also differed significantly from the Forgery label on all four dimensions (Beauty: -3.4% [-4.5, -2.3]; Valence: -2.7% [-3.6, -1.7]; Meaning: -6.2% [-7.6, -4.8]; "$0" option: +8.8% [7.6, 10.1]). These effects are in line with the preregistered predictions of lower meaningfulness and worth for AI-Generated images and of lower beauty and worth for Forgeries, and extend them to all four dimensions. For meaningfulness, the labels additionally increased the probability of a flat "not at all meaningful" (0) answer, markedly so for AI-Generated images (+4.9% [3.1, 7.4]; Forgeries: +0.9% [0.2, 1.8]). For beauty, both stages of the judgment were affected: the labels lowered the probability of judging an image as beautiful rather than ugly (Forgeries: -6.8% [-9.6, -4.0]; AI-Generated: -15.9% [-19.3, -12.5]) as well as the degree of beauty reported among the images judged beautiful, while images judged ugly were judged more decidedly so. None of the gaze features (entropy, laterality, centeredness, maximal shift) differed between label conditions.

Regarding the affective properties of the artworks, their normative emotion category shaped the ratings independently of the label. Relative to positive, low-intensity artworks, negative, high-intensity artworks were rated as less beautiful (-23.8% [-29.2, -17.0]) and less pleasant, but as more meaningful (+11.7% [5.3, 18.3]), and were more often assigned no monetary value; negative, low-intensity works showed the same pattern without the gain in meaningfulness, and intensity made little difference among positive works.

**Figure 2**

*Effect of the label condition on the four Phase 1 aesthetic judgments. Grey bars and histograms show the observed distribution of ratings; coloured lines and point-ranges show the model-implied response distribution for each label condition with its 95% Credible Interval.*




The label effects, however, did not depend on the kind of artwork: within each of the four emotion categories, all Phase 1 rating contrasts between the three labels remained significant and in the same direction, with comparable magnitudes.

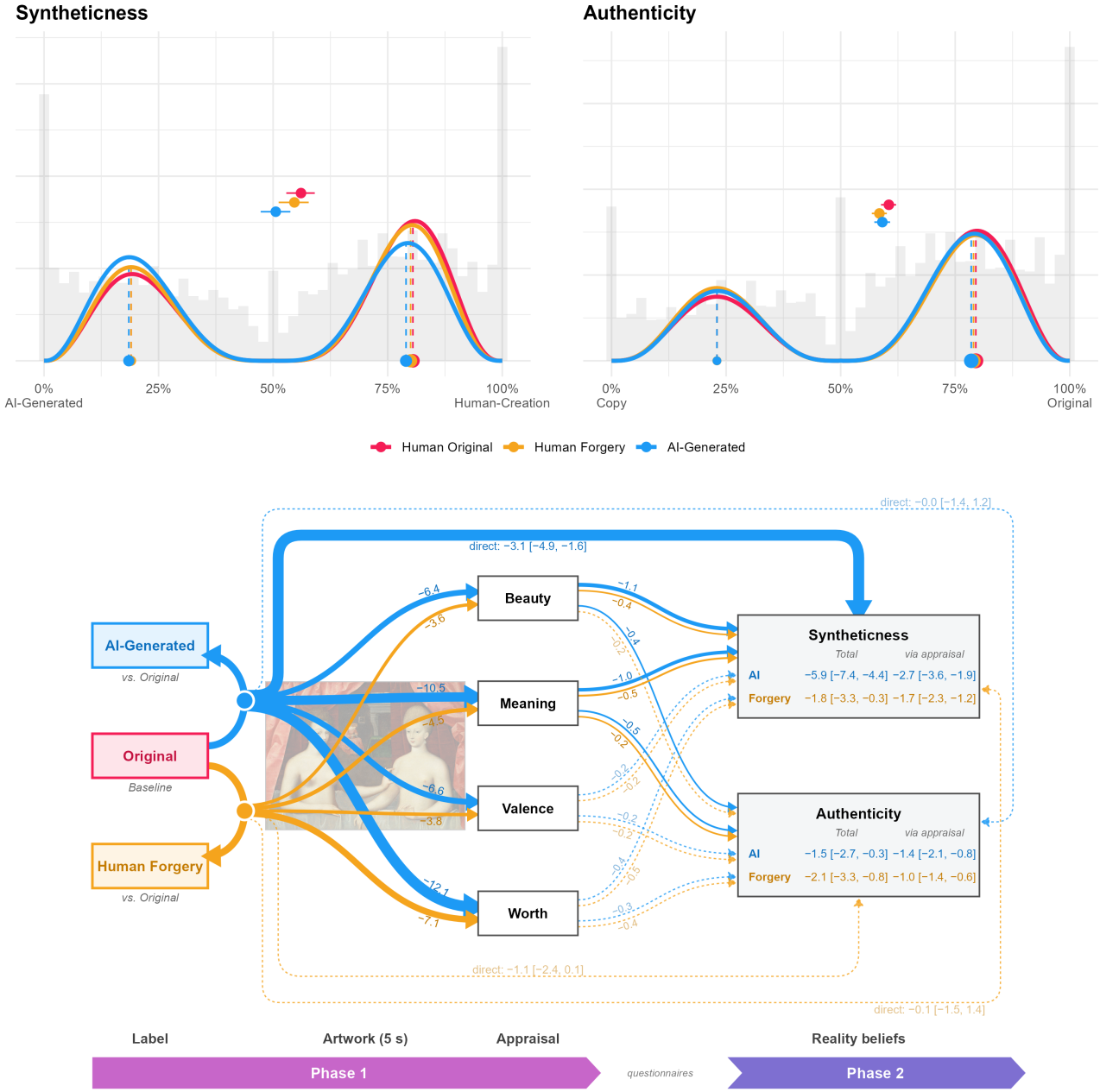
## Syntheticness and Authenticity Beliefs

The labels also shaped the reality beliefs measured in Phase 2 in a dissociable manner (Figure 3, top). As preregistered, images that had been presented as AI-Generated were subsequently judged as more synthetic, i.e., less likely to be human-made, than Originals (-5.5% [-6.9, -4.1]), and also than Forgeries (-4.0% [-5.5, -2.5]), whereas the Forgery label barely affected syntheticness (-1.5% [-2.9, -0.1]). Conversely, the Forgery label lowered judged authenticity relative to Originals (-2.0% [-3.3, -0.9]), and so, to a smaller extent, did the AI-Generated label (-1.4% [-2.6, -0.2]); the two "fake" labels did not differ from each other (+0.7% [-0.6, 2.0]). In other words, the AI label was encoded specifically as a matter of syntheticness, whereas the Forgery label produced smaller and less specific shifts, lowering authenticity somewhat more than the AI label did but not significantly so. Within each emotion category, the effect of the AI label on syntheticness remained significant, whereas the smaller effects were less precisely estimated and reached significance in at most two categories. That the labels still informed the beliefs is expected to some extent: participants had been told that only *some* of the labels had been randomized, so drawing on a label as a partially valid cue was a reasonable use of the information rather than a persistence of belief against correction.

Since the labels lowered both the aesthetic judgments of Phase 1 and the reality beliefs of Phase 2, we examined whether the latter effect was carried by the former, i.e., whether artworks were judged as more synthetic or less original *because* the label had made them look less beautiful. As preregistered, beauty was positively associated with both beliefs: artworks rated as more beautiful (relative to the participant's other ratings) were judged as more likely to be human-made (+3.4% [2.8, 4.0] per 10% increase in beauty, for Originals) and as more original (+1.9% [1.5, 2.4]), a relationship slightly weaker under the two "fake" labels.

**Figure 3**

*Effect of the label condition on reality beliefs, and its decomposition through the Phase 1 appraisal. Top: Phase 2 syntheticness (0% = AI-Generated, 100% = Human creation) and authenticity (0% = Copy, 100% = Original) judgments. Grey histograms show the observed ratings; coloured lines show the model-implied response distribution per label condition, dots on the x-axis the modal rating on each side of the scale (sized by the probability of choosing that side), and point-ranges the marginal means with their 95% CI. Bottom: decomposition of the label effects (blue: AI-Generated vs. Originals; orange: Forgeries vs. Originals) through the four Phase 1 ratings, laid out on the timeline of the experiment. Label → rating paths give the effect of the label on each rating (% of its scale); rating → belief paths give the unique indirect effect carried by that rating; the outer paths give the direct effect of the label on each belief; each belief box gives the total effect and the joint indirect effect (via appraisal) with their 95% CI. Effects on beliefs are in % of the belief scale. Line width scales with the size of the effect; dashed, faded paths have a 95% CI including 0.*




Decomposing the label effects into a direct and an indirect (beauty-mediated) component revealed a dissociation, clearer for the AI-Generated label than for the Forgery label. The AI-Generated label lowered syntheticness mostly directly (direct effect: -4.0% [-5.5, -2.6]), with about a third of its total effect (30% [20, 41]) carried by the lower beauty it induced (indirect effect: -1.7% [-2.2, -1.3]), whereas the small effect of the Forgery label on syntheticness was fully

accounted for by beauty (direct: -0.7% [-2.0, 0.7]; indirect: -1.0% [-1.3, -0.7]). The mirror pattern emerged for authenticity: the Forgery label lowered authenticity mostly directly (direct: -1.5% [-2.7, -0.2]; indirect: -0.5% [-0.7, -0.3]), whereas the effect of the AI-Generated label was carried by beauty (direct: -0.6% [-1.8, 0.7]; indirect: -0.8% [-1.2, -0.5]). Each label thus directly shifted the belief dimension it semantically implied - algorithmic origin for AI-Generated images, lack of originality for Forgeries - whereas its spillover onto the other dimension was carried by the less favourable appraisal it had induced. These direct effects operated on the categorical choice (human vs. AI, original vs. copy) rather than on the confidence of the judgments. The Forgery side of this pattern rests on a smaller and less robust direct effect: authenticity rose with beauty only above the participant's average (beautiful works were taken as originals, but ugly ones were not taken as copies), and allowing for this curvature, although it did not improve predictive accuracy, made the direct effect of the Forgery label marginal (-1.3% [-2.7, 0.2]).

When valence, meaningfulness and worth were entered as mediators alongside beauty (Figure 3, bottom), the four ratings jointly carried almost half of the effect of the AI-Generated label on syntheticness (indirect: -2.7% [-3.6, -1.9]; direct: -3.1% [-4.9, -1.6]), the whole of the two spillover effects, and about half of the effect of the Forgery label on authenticity. The direct effect of the Forgery label on the mean authenticity rating was then no longer credible, but the label still directly lowered the probability of judging an artwork as original rather than as a copy (-2.2% [-4.5, -0.1]). Within the joint indirect effects, only beauty and meaningfulness made unique contributions, as preregistered for meaningfulness, whereas valence and worth did not, although the labels lowered them at least as much as beauty. The dissociation between the two labels therefore holds in relative rather than absolute terms: the Forgery label lowered authenticity both through the appraisal it induced and directly, by tipping the categorical choice between original and copy, whereas the AI-Generated label did so only through the appraisal.

Because beauty was not manipulated, these indirect effects could also reflect a common cause, such as a participant's label-independent preference for particular artworks shaping both their beauty and their reality judgments. Adding the beauty rating given at follow-up (after the debrief, and unaffected by the labels) as a covariate left the pattern of direct and indirect effects unchanged. Follow-up beauty had its own, additive association with syntheticness, suggesting that reality beliefs drew both on how much participants liked an artwork regardless of its label and on the label-induced change in their appraisal of it.

Finally, the association between beauty and perceived human origin did not require any label. At follow-up, the more beautiful participants found an artwork that they judged as new, the less likely they rated it to be AI-generated, with the same slope for genuinely new artworks, which had never been associated with a label (-2.2% [-2.7, -1.7] per 10% increase in beauty), and for previously seen artworks that were not recognized (difference: 0.0% [-0.5, 0.5]). We describe this association as a "beautiful = human" inference, but its direction is not fixed by the design: beauty was rated before the belief in both sessions, yet an impression that a work "looks artificial" can form during viewing and lower its beauty, so the same association would arise from a spontaneous, label-free anti-AI bias operating at the level of the artwork. The follow-up beauty control rules out stable preferences for particular artworks as a common cause, not a stable impression of their artificiality.

Properties of the artworks themselves also informed syntheticness judgments, independently of the label: negative, high-intensity works, works of higher normative arousal, and Abstract and Avant-garde and Romantic and Realism works (compared with Classical and Impressionist and Expressionist ones) were judged as more synthetic. Normative liking did not predict syntheticness, nor did normative valence, complexity or familiarity, so that the "beautiful = human" inference relied on the participant's own appraisal of an artwork rather than on its normative appeal. Authenticity varied little with any of these properties.

## The Role of Self-Relevance

Self-relevance ratings were skewed toward the low end of the scale (38.5% of the artworks were rated as "not at all" self-relevant) and were not affected by the label under which an artwork had been presented in Phase 1. Within participants, self-relevance was most strongly related to the beauty rated in the same session ($r$ = .58) and to Phase 1 beauty (.40), and only weakly to meaningfulness and to the reality beliefs (≤ .17).

Artworks that participants found more self-relevant at follow-up had been rated as more beautiful (+3.2% [2.6, 3.6] per 10% increase in self-relevance, for Originals) and more meaningful in Phase 1, similarly under the three labels. The label penalty, however, did not change with self-relevance. The beauty gap between AI-Generated images and Originals was of the same size for artworks more self-relevant than the participant's average as for less self-relevant ones (+1 SD: -6.3% [-8.1, -4.7]; -1 SD: -6.5% [-8.3, -4.6]; difference: +0.2% [-2.3, 2.5]). The same held for the gap between Forgeries and Originals, for meaningfulness, with follow-up beauty as a competing moderator, and for the categorical components of the judgments (the probability of judging an image as beautiful, and of rating it as "not at all meaningful"), whose label effects were constant on the odds scale. Nor did the penalty depend on how self-relevant participants found the artworks overall: participants with a higher mean self-relevance rated the artworks as more beautiful, but their beauty gap between AI-Generated images and Originals was not significantly smaller.

Self-relevance also contributed, modestly, to reality beliefs, over and above both beauty ratings: more self-relevant artworks were judged as more likely to be human-made under the AI-Generated label (+1.0% [0.4, 1.6] per 10% increase; the difference from the other labels was not significant) and as more original under the two "fake" labels.

These contributions left the slope of Phase 1 beauty virtually unchanged, so that the "beautiful = human" inference was not a proxy for personal resonance, and, as self-relevance was unaffected by the labels, it carried none of the label effects on the beliefs. As for beauty, this link did not require any label: at follow-up, the more self-relevant an artwork judged as new, the less likely it was rated to be AI-generated, over and above its beauty, for new and unrecognized old artworks alike. Self-relevance, rated in the same session, absorbed about a quarter of the association between follow-up beauty and perceived human origin.

## Persistence and Memory

### *Persistence of the Label Effect*

The label effect was no longer detectable about six weeks later, at the follow-up session, where participants had been reminded that all artworks were genuine human originals: artworks previously labelled as AI-Generated or as Forgeries were not rated as less beautiful than those labelled as Originals (-0.9% [-2.0, 0.1] and -0.3% [-1.4, 0.9], respectively; about a tenth of the Phase 1 effects; Figure 4, bottom left), and the original label affected neither the perceived artificiality of the artworks that participants did not recognize nor, as reported above, their self-relevance. This design cannot separate the passage of time from the explicit correction, or from the fact that only a subset of participants returned (220 of 324).

Two further models looked for what this null might hide. First, a residual label effect appeared only among the artworks that participants still recognized (43.1% of those seen in Phase 1), and only on the categorical component of the judgment: recognized artworks previously labelled as AI-Generated were somewhat less likely to be placed on the "beautiful" side of the scale than recognized Originals (-4.6% [-9.1, -0.3]), whereas unrecognized ones were not. Second, the syntheticness belief formed in Phase 2 carried into the follow-up: given the Phase 1 beauty of an artwork, the more human-made a participant had judged it to be, the more beautiful they found it six weeks later, so that the AI label retained a small indirect effect on follow-up beauty through the belief it had induced (-0.3% [-0.4, -0.1]). The trace that the label left in the aesthetic judgment itself was thus small and confined to the artworks that participants still recognized; it was clearer in the belief about the artwork, which in turn still coloured its later appreciation.

### *Memory for Labels and Beliefs*

Participants recognized 43.1% of the artworks seen in Phase 1, against a false-alarm rate of 11.1% for new artworks. Recognition did not depend on the label under which an artwork had been presented, nor on the participant's own Phase 2 belief about it, but on how the artwork had been appraised in Phase 1. In line with our previous work linking extreme aesthetic judgments to a mnemonic advantage (Lee et al., 2023; Salgues et al., 2024), we had hypothesized that artworks receiving extreme ratings, whether positive or negative, would be better remembered. This held for the positive extreme only: artworks rated as more beautiful than the participant's average (and, as the two ratings usually went together, as more positive) were more likely to be recognized, increasingly so toward the top of the range (+12.0% [6.9, 17.3] at two SD above the participant's mean; Figure 4, bottom), whereas artworks rated as less beautiful than average were not (+0.0% [-4.1, 4.2] at two SD below). Of the two ratings, only beauty retained a credible effect when each was varied with the other held constant. The artworks with the highest recognition rates were nonetheless the ugliest and most negative ones, but this was a property of the artworks (those of higher normative arousal, and those found uglier on average, were better recognized by everyone), which the model separates from each participant's own appraisal: within a given artwork, it was finding it more beautiful than usual, not finding it extreme, that made it more memorable.

Self-relevance, rated at follow-up, was also related to recognition, but not in the way a self-reference effect would predict. More self-relevant artworks were more often reported as "seen before", but as much when they were new as when they had actually been seen (Figure 4, bottom right): the effect was larger in percentage points for old artworks only because hits were far more frequent than false alarms, and on the odds scale it did not differ between old and new artworks (difference in log-odds ratio at two SD above the mean: -0.11 [-0.38, 0.16]). The same held for follow-up beauty. Self-relevance and beauty at follow-up thus indexed a bias to judge a resonant or appealing artwork as familiar rather than better memory for it, which is why the Phase 1 appraisal (rated before any memory judgment, and only for artworks actually seen) is the cleaner index of what was encoded.

Memory for the label itself was at chance: artworks were no more often attributed the label they had actually carried than any other label (all differences ≤ 1.3 percentage points, all CIs including 0). Instead, label attributions followed a stable response bias: whatever their actual label, recognized artworks were most often remembered as being associated with the "Original" label (51% of recognized artworks), followed by "AI-Generated" (33%) and "Forgery" (15%). New artworks that were falsely recognized received the same distribution of labels (53%, 32% and 15%), suggesting that label attributions were reconstructed rather than retrieved. The preference for the Original label may be partly driven by the reminder, at the start of the follow-up session, that all artworks were genuine human originals.

In contrast, participants' recollection of their own Phase 2 belief was above chance, mainly along the syntheticness dimension (Figure 4, top). Artworks that a participant had judged to be AI originals were more often recalled as such than artworks judged to be Human Originals (+6.5% [4.7, 8.5]), and less often recalled as Human Originals (-8.0% [-9.9, -6.2]), whereas artworks judged to be Forgeries were not more often recalled as Forgeries than those judged to be Originals, nor were AI copies more often recalled as AI copies. As the model included participant- and item-

**Figure 4**

*Memory for the labels and for one's own beliefs. Top: the artworks recognized at follow-up (4,497 from 214 participants). In the middle, each artwork's path from the label recalled for it (left), through the participant's own Phase 2 belief about it (middle; dichotomized syntheticness and authenticity ratings), to the belief recalled for it (right); the boxes give the percentage of artworks in each category and the flows are coloured by the Phase 2 belief, so that a flow ending in a box of its own colour marks an answer consistent with the belief; the numbers at each end of a flow give its share of the box it touches (next to the Phase 2 belief, the percentage of the artworks with that belief given each answer; next to the answer, the percentage of the artworks with that answer coming from each belief). On either side, the same data as matrices: for the artworks of each own belief (columns, each summing to 100%), the percentage recalled with each label (left) or belief (right); outlined cells: the answer consistent with the belief. The label shown in Phase 1 is not represented, as it predicted none of the follow-up answers. Bottom, from left to right: beauty under each label in Phase 1 and at follow-up (marginal means with their 95% CI); the probability of recognizing an artwork by its Phase 1 beauty; the label recalled for recognized artworks by their Phase 1 beauty; and the probability of judging an artwork as seen before by its self-relevance rated at follow-up, for old (hits) and new artworks (false alarms). Ratings are expressed in points above or below the participant's own mean; curves are population-level predictions with their 95% CI.*




level variability in each answer, this correspondence is not a matter of general response tendencies, or of some artworks looking more artificial than others to everyone.

This correspondence may, however, reflect a stable, personal impression of an artwork re-expressed at follow-up as much as an episodic memory of the past judgment: beauty and self-relevance were rated again immediately before the memory questions, and beauty was closely associated with the belief in Phase 2. It nonetheless survived when the current impression was taken into account: with the Phase 1 beauty, follow-up beauty and self-relevance of the artwork in the model, the belief effects were unchanged (artworks judged to be AI originals recalled as such: +7.1% [5.2, 9.0]), although the current impression shaped the answers on its own, artworks currently found more beautiful being more often recalled as Human Originals (+10.7% [6.8, 15.3] of recognized artworks at two SD above the participant's mean), as were, to a lesser extent, more self-relevant ones. The recalled belief was thus both a memory of the Phase 2 judgment and a re-inference from the present appraisal, the two contributing independently.

The recalled label, in turn, appeared to be reconstructed from this belief and from the Phase 1 appraisal (Figure 4, top and bottom). With the label actually shown held con-

stant, artworks that a participant had judged to be AI originals were more often attributed the "AI-Generated" label than those judged to be Human Originals (+7.0% [5.0, 8.9]; among recognized artworks: 43.6% vs. 27.3%), again along the syntheticness dimension only (artworks judged to be Forgeries were not more often attributed the "Forgery" label). The "beautiful = human" inference described above also entered this reconstruction: artworks rated as less beautiful than the participant's average in Phase 1 were more often remembered as "AI-Generated" (+7.4% [1.8, 12.7] of recognized artworks at two SD below the mean) and those rated as more beautiful more often as "Original", whereas the "Forgery" attribution did not follow beauty. This effect was smaller, but still present, with the belief, follow-up beauty and self-relevance held constant, and it was the Phase 1 appraisal rather than the current one that mattered, follow-up beauty having no effect. Self-relevance acted in the same direction, more self-relevant recognized artworks being more often attributed the "Original" label and less often the "AI-Generated" label. In other words, participants remembered what they had come to believe about an artwork, and what they had felt about it in Phase 1, but not what they had been told about it.

## Interindividual Differences

Participants differed reliably in their baseline responses, but much less in their sensitivity to the labels. All baseline indices were reliable (D-vour from .58 to .97), whereas individual differences in the label effects were mostly dominated by estimation uncertainty: the effect of the AI-Generated label was moderately reliable on the Phase 1 ratings (.34 to .71), the effect of the Forgery label only on worth (.55), and none of the label effects on the Phase 2 beliefs was (all ≤ .13). With 16 artworks per label and participant, the latter result does not show that participants did not differ in how the labels shifted their beliefs, but that such differences, if any, were too small to be measured with this design. The "beautiful = human" slope was moderately reliable (.44).

The 28 baseline and label-effect indices with a reliability above 1/3 formed nine lower-order communities, grouped into three higher-order communities (Figure 5, top). The first, *aesthetic appraisal*, gathered the Phase 1 indices into four communities: *aesthetic sensitivity*, the baseline level of the four ratings (finding the Originals beautiful, pleasant, meaningful and valuable); *anti-fake bias*, the effects of the AI-Generated label on the four ratings and of the Forgery label on worth; *rating consistency*, the precision of the four ratings; and *ugliness extremity*, how far into the "ugly" side of the beauty scale participants went when they chose that side. The anti-fake bias thus formed a dimension of its own, largely independent of how much participants appreciated the artworks in general ($r$ = .17). The Phase 2 indices formed the two other higher-order communities: *realness beliefs*, comprising the baseline tendency to judge the artworks as human-made and original (*realness tendency*) and the extremity of these judgments on the human / original side (*realness conviction*) and on the AI / copy side (*fakeness conviction*), and *belief consistency*, comprising the precision of the judgments on either side. Each Phase 2 community paired a syntheticness index with its authenticity counterpart: the extremity and precision of the two beliefs were strongly correlated, whereas the tendencies to judge the artworks as human-made and as original were only weakly related, so that participants differed in a shared way of using the two belief scales rather than in a general belief that the artworks were fake. As each higher-order score was dominated by one or two of its lower-order communities ($r$ = .97 to .99), we report the correlates of the lower-order communities only. This structure was moderately stable: across 500 bootstrap samples, three higher-order communities were found in 56% of the samples, and nine lower-order communities was the most frequent solution (31%).

Two further sets of indices were kept out of this structure. The "beautiful = human" slope was nearly unrelated to all other indices and to all participant characteristics (all $|r| \leq$ .18). At follow-up, the tendencies to attribute each label to recognized artworks were reliable individual characteristics (D-vour .69 to .80), unlike the accuracy of these attributions (.09), but they were unrelated to all Phase 1 and Phase 2 indices (all $|r| \leq .17$): how inclined a participant was to attribute each label was unrelated to how they had, overall, appraised the artworks and judged their origin (at the level of the artworks, by contrast, the attributions followed the appraisal and the belief; see Memory for Labels and Beliefs).

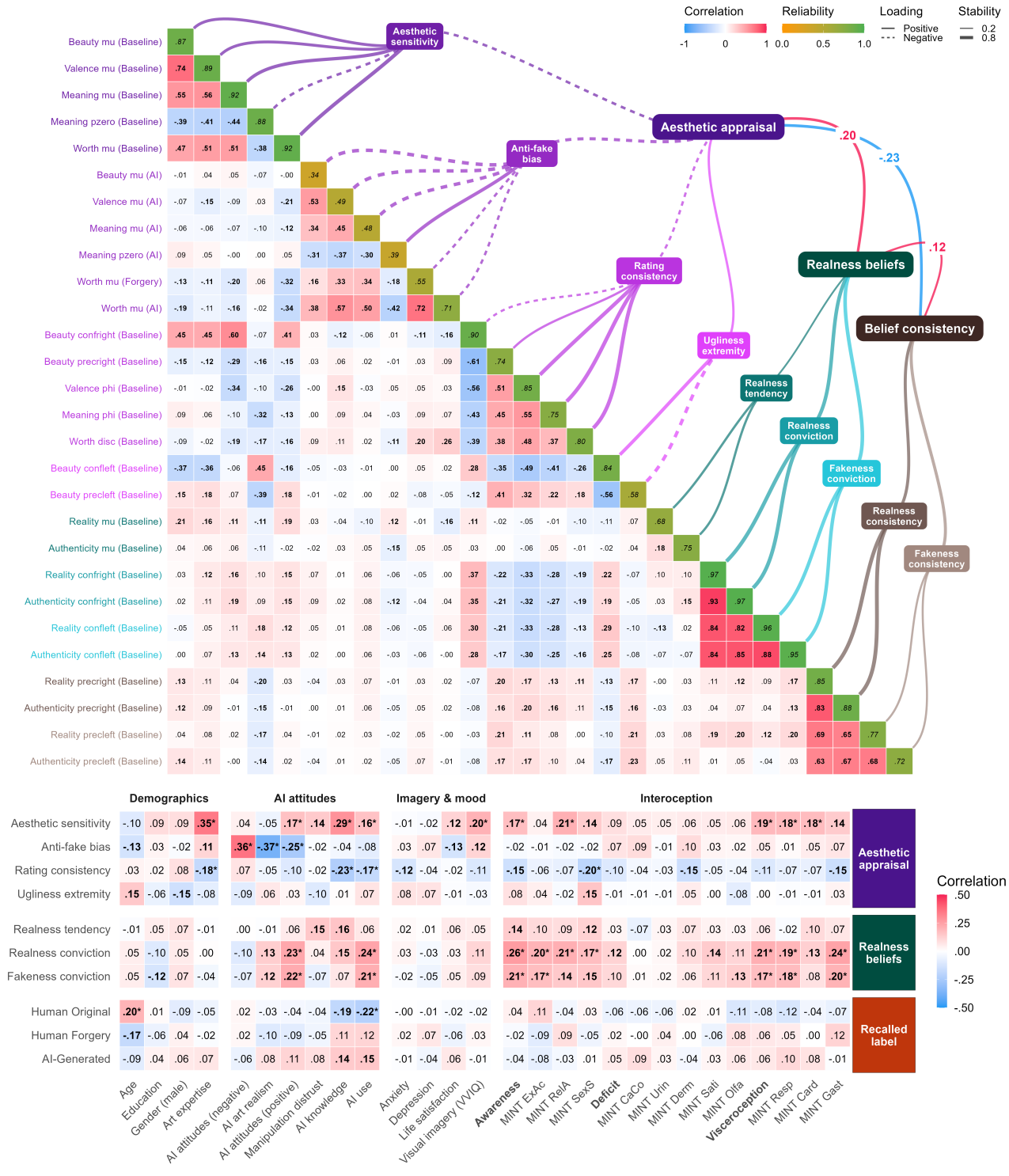
## Dispositional Correlates

The community scores and the label recall tendencies were then correlated with the participants' characteristics (Figure 5, bottom). As preregistered, the anti-fake bias was related to attitudes toward AI, and to no other characteristic after correction: participants with more negative attitudes toward AI penalized the non-original labels more strongly ($r$ = .36), whereas those who believed that AI-generated art can rival human art ($r$ = -.37) and those with more positive attitudes toward AI ($r$ = -.25) penalized them less. This association was carried by the part of the penalty specific to the AI label: the participant-level effect of the Forgery label was largely unrelated to attitudes toward AI, whereas the increment from the Forgery to the AI label was related to all three attitude scales. Contrary to the preregistered predictions, the bias was not larger in participants with higher interoceptive sensibility (all $|r| \leq .10$), and its association with more vivid visual imagery was weak ($r$ = .12) and did not survive the correction. The bias was also unrelated to manipulation distrust, art expertise, and knowledge and use of AI tools.

Interoceptive sensibility was instead related to the baseline level of the judgments: participants reporting more interoceptive awareness and visceroception found the Originals more beautiful, pleasant, meaningful and valuable, as did participants with more art expertise, more knowledge and use of AI, more positive attitudes toward AI and more vivid visual imagery. Interoceptive sensibility also went

**Figure 5**

*Structure and correlates of the participant-level indices. Top: correlations between the 28 indices with a reliability (D-vour) above 1/3 (lower half; bold:* p < *.05), their reliability (diagonal) and their hierarchical structure (upper half; edge width: item stability over 500 bootstrap samples; dashed: negative loading; arcs: correlations between the higher-order scores); the columns follow the order of the rows. Baseline: Originals; Forgery, AI: difference from Originals; mu: location; confright / confleft and precright / precleft: extremity and precision of the judgments on the right (beautiful, human, original) and left side of the scale; phi, disc: precision; pzero: probability of a 'not at all meaningful' answer; Reality: syntheticness. Scores are oriented so that a higher score means more of what the community label says (e.g., a larger anti-fake bias), hence the negative loadings of the label effects. Bottom: correlations between the lower-order community scores and the tendencies to attribute each label to recognized artworks at follow-up (rows) and the participants' characteristics (columns); bold:* p < *.05 (uncorrected); asterisks:* p < *.05 after false-discovery-rate correction across the 280 correlations; colour scale truncated at* ±*.5.*



with more decisive reality beliefs: participants reporting more interoceptive awareness made more extreme syntheticness and authenticity judgments, both on the human / original side (realness conviction: $r$ = .26) and on the AI / copy side (fakeness conviction: $r$ = .21), as did those reporting more visceroception, using AI tools more often and holding more positive attitudes toward AI. This association was specific to the reality judgments, which participants had been asked to make on "gut feeling" (interoceptive sensibility was unrelated to the extremity of ugliness judgments), and may reflect a greater reliance on, or confidence in, felt impressions. Finally, participants with more art expertise and more knowledge and use of AI tools rated the artworks less consistently from one artwork to the next. The ugliness extremity and realness tendency scores had no correlate surviving the correction.

At follow-up, the tendency to attribute the "Original" label to recognized artworks was higher in older participants ($r$ = .20) and lower in participants who used AI tools more often ($r$ = -.22). These associations are small, and survived the correction only as part of the whole set of correlations, but they are consistent with the reconstructive nature of label memory (see Memory for Labels and Beliefs): in the absence of a memory of the label, participants' attributions may follow their prior expectations about how common AI-generated images are.

## Discussion

Is art devalued when attributed to AI because it is not an authentic original, or because it was not made by a human at all? By presenting the same genuine paintings as Originals, Forgeries or AI-Generated, we separated these two components and followed the bias from the moment of judgment to what remained of it six weeks later. Forgeries were judged less beautiful, less pleasant, less meaningful and less valuable than Originals, and AI-Generated works about twice as much so, whatever the emotional content of the painting and without any change in how it was looked at. Asked afterwards about each artwork's origin, participants judged those labelled AI-Generated as more synthetic and, less clearly, those labelled Forgeries as less original. These beliefs also followed participants' own aesthetic appraisal: the more beautiful they had found an artwork, the more likely they were to judge it human-made and original, and the link with human origin held even for newly presented artworks. Six weeks later, participants no longer showed the bias and forgot the labels. What they remembered, and attributed to the label, was what they had believed and felt.

### Not All Fakes Are Equally Ugly

The intermediate position of Forgeries shows that the anti-AI bias is partly a generic penalty for non-original works and partly specific to algorithmic authorship. A human copy was devalued on every dimension, as contagion accounts predict for an object that lacks the trace of its author (Huang et al., 2011; Newman & Bloom, 2012). The AI-specific component is not, however, a stronger version of the non-originality penalty: the AI label lowered judged authenticity no more than the Forgery label, and only through the poorer aesthetic appraisal it induced, whereas it made the artworks look more synthetic directly. What the AI label adds is the absence of a human author, in line with accounts that tie the value of a work to the human effort (Kruger et al., 2004) and intentions (Bullot & Reber, 2013; Jucker et al., 2014) it is taken to embody, both of which a forger retains and an algorithm lacks.

This added penalty was the heaviest for meaning. Meaningfulness was the rating the AI label lowered most, the only one for which the increment from the Forgery to the AI label exceeded the penalty the two labels shared, and the only one the AI label drove to outright denial (answering the lower extreme - "0", to the meaning scale), a response the Forgery label barely touched. This matches the observation that the anti-AI bias is largest for judgments that attribute meaning, intention or depth to a work (de Rooij, 2025), and the idea that meaning presupposes an agent whose intentions the work expresses: a forgery reproduces the content, and with it some of the meaning, that an artist put into the original, whereas a work attributed to an algorithm can be denied meaning altogether. In the terms of Leder et al. (2004), the label acts at the stages of cognitive mastering and evaluation, where a work is interpreted in light of its presumed author, not at perceptual analysis. Consistent with this, the labels left visual exploration unchanged, as they left physiological responses unchanged in Chiarella et al. (2022), and as expected if the bias grows under reflective processing (Wang et al., 2026). The gaze differences reported by Zhou and Kawabata (2023) arose between paintings that participants had themselves classified as human- or AI-made, and may reflect the visual properties that drove the classification, which a randomly assigned label does not carry. Webcam eye tracking is coarse, and although its features did track the ratings (see Methods), this absence of effect should be interpreted with caution.

Is there such a thing as an anti-AI bias? Our results indicate about half of what the AI label cost was also paid by a human forgery, a penalty for works that are not their author's own, documented long before AI (Newman & Bloom, 2012); the other half was specific to the absence of a human author and was encoded as the dimension of "syntheticness" rather than as a further lack of authenticity. The devaluation is thus better described as an "authorship premium", which likely draws from two sides of authorship, the making of a work and its maker. The making, the effort, skill and creative conception a work is believed to embody, mostly depends on there being a maker, not on who that maker is. The maker, in turn, is a source from whom a work inherits the regard in which they are held, as objects do from contact with admired or despised people (Newman et al., 2011) and paintings from the name attached to them (Kaube & Abdel Rahman, 2024; Mastandrea & Crano, 2019), and the penalty specific to AI did grow with negative attitudes toward it. The converse manipulation, varying the making while holding the maker constant, suggests that the making counts only when credited to a mind: making an AI take longer to

generate an image led people to credit it with more effort, but not to appreciate the image more, appreciation following instead their attitudes toward AI and the consciousness they attributed to it (Theirs et al., 2026). Varying the maker while holding the making constant would complete the picture: if the same image were rated higher when attributed to a new, acclaimed model than to an older, less regarded one, it would confirm that part of what is called an anti-AI bias is a contagion from a disliked maker rather than a judgment about how the work was made.

## Affect Shapes the Judgment but Not the Bias

The affective profile of an artwork shaped the judgments but not the bias. Negative, high-intensity works were judged far less beautiful and pleasant yet more meaningful, and the label effects were nonetheless present, in the same direction and of similar size, in every emotion category: the labels added a penalty on top of a work's affective profile rather than acting through it. Nor was there any sign that the AI label acted as a cue of unreality that dampens the emotional response, as in fictional reappraisal (Makowski et al., 2019; Sperduti et al., 2016). Such dampening would have made negative works less unpleasant under the AI label; instead, the labels lowered valence in every category, negative works included. Self-reported valence, which earlier studies found insensitive to authorship (Demmer et al., 2023; Zhou & Kawabata, 2023), was thus affected here, but as a downward shift comparable to that of beauty, not as a blunting of the response. This does not rule out a dampening, however. Within an artistic frame, even scenes of martyrdom, war or death are depictions held at a distance, whose negative emotions can be savoured rather than avoided (Menninghaus et al., 2017), and the valence ratings, although the question asked how the artwork made participants feel, may have recorded how sad or dark a work was more than an unpleasant feeling that a cue of unreality would down-regulate. Whether an AI label dampens the response to genuinely aversive material, such as the distressing scenes used in research on fictional reappraisal, remains to be tested. The drop in valence under the fake labels may likewise reflect a judgment about the artwork more than a change in how it made participants feel, in line with the absence of physiological effects of an AI label (Chiarella et al., 2022) and with a bias that operates through explicit, deliberate judgment (Bara et al., 2025).

## Labels, Appraisal, and Beliefs About Origin

The two fake labels were not encoded as one undifferentiated sense of fakeness. The AI label made artworks look more synthetic than both Originals and Forgeries, mostly directly. The Forgery label had a smaller effect on authenticity, not significantly larger than that of the AI label, but the only one with a direct component. Each label thus shifted directly the belief dimension it semantically implied, and spilled over onto the other dimension only through the poorer appraisal it induced; both direct effects acted on which side of the scale participants chose, not on their confidence within that side. The dissociation is asymmetric: the AI label was a strong and specific cue, the Forgery label a weak one. A copy has no perceptible signature when the original is unknown, whereas people hold expectations about what AI images look like. Accordingly, the style and normative arousal of the artworks predicted syntheticness, whereas nothing predicted authenticity, and across participants the two beliefs shared a response style but not a content: those inclined to judge the artworks human-made were not particularly inclined to judge them original. Because participants had been told that only some labels had been randomized, these effects reflect the sensible use of a partially valid cue rather than the persistence of a belief against its correction.

Aesthetic appraisal, in turn, informed belief. As preregistered, the more beautiful participants found an artwork, the more likely they judged it human-made and original, and this association needed no label: at follow-up, the more beautiful an artwork judged as new, the less likely it was rated AI-generated, including for new artworks that had never been labelled. The same inference has been observed for faces, attractive faces being less often judged artificial (Makowski, Te, Neves, Kirk, et al., 2025), and it extends to provenance the halo by which the beautiful is assumed to be good (Dion et al., 1972). It has two consequences. Under a label, it closes a loop: the label lowers the appraisal, and the lower appraisal confirms the artificial origin, with about a third of the effect of the AI label on syntheticness manifesting this way. Without a label, appeal itself becomes evidence of origin, so that appealing AI images may pass as human-made, as AI-generated faces do (Miller et al., 2023; Nightingale & Farid, 2022), and human works that fail to please may be dismissed as AI-generated. The direction of the association remains an interpretation: an impression that a work looks artificial can form during viewing and lower its beauty, and a spontaneous, label-free anti-AI bias at the level of the artwork would produce the same pattern. Controlling for follow-up beauty excluded stable preferences for particular artworks as a common cause, not stable impressions of their artificiality. On either reading, aesthetic appraisal and beliefs about provenance are entangled for images of unknown origin, and the inference rests on the viewer's own appraisal rather than on a work's normative appeal, which did not predict syntheticness.

Of the four ratings, only beauty and meaningfulness carried unique parts of the label effects on beliefs; valence and worth, although lowered as much, did not. The route from appraisal to belief thus runs through evaluative and semantic judgment rather than through affect or economic value. Meaningfulness was both the rating the AI label depressed most and one of the two carrying its effect on belief, which suggests a two-way link between attributing meaning and attributing a human author: the AI label leads viewers to deny a work meaning, and a work that feels meaningless is taken to be AI-generated.

## Self-Relevance: A Cue to Human Origin, Not a Shield Against the Label

Self-relevance did not moderate the bias. Artworks that participants found more self-relevant had been judged more beautiful and meaningful, but the gap between labels was the same for them as for less self-relevant ones, within and between participants and for the categorical components of the judgments alike. Synthetic images personalized around a viewer's own memories and identity can be as appealing as human art (Vessel et al., 2023); personal resonance with a given work, however, did not protect it from the devaluation that follows from attributing it to an algorithm or a forger. Two features of the design temper this conclusion. The artworks were not selected for personal relevance, and many were rated as not self-relevant at all, so that the autobiographical resonance produced by personalization was rarely reached; and self-relevance was rated at follow-up, after the debrief, as a stable property of the relation between viewer and work rather than while the label was present.

Self-relevance nonetheless played a role of its own in beliefs about origin. Beyond both beauty ratings, more self-relevant artworks were judged more human-made under the AI label and more original under the two fake labels, were less likely to be rated AI-generated when judged new at follow-up, and were more often remembered as Originals than as AI-Generated. Viewers appear to infer that a work that speaks to their own experience was made by someone with experiences of their own, a "self-relevant = human" inference that complements the "beautiful = human" one. In memory, by contrast, self-relevance produced no self-reference effect: it raised "seen before" answers as much for new as for old artworks, a familiarity bias rather than better memory. Unlike in studies that assessed self-relevance at encoding (Lee et al., 2023; Salgues et al., 2024), it was rated here at retrieval, where it may itself feed on a feeling of familiarity. The appraisal made at encoding did predict recognition, but on the positive side only: artworks rated as more beautiful than the participant's average were better remembered, those rated as less beautiful were not, which only partly supports a mnemonic advantage of extreme aesthetic judgments.

## What Remains of the Label

About six weeks later, reminded that all artworks were human originals, participants no longer rated previously AI- or Forgery-labelled artworks as less beautiful, more artificial or less self-relevant. The effect on beauty fell to about a tenth of its Phase 1 size. The bias thus behaves mainly as an online inference from the information at hand, not as a lasting encoding of the work, and in this differs from the perseverance of beliefs after debriefing (Ross et al., 1975) and from the continued influence of corrected misinformation (Lewandowsky et al., 2012), at least as far as the aesthetic judgment itself is concerned. Because time, correction and attrition are confounded in this design, the fading cannot be credited to the correction.

In the follow-up ratings, what persisted was the belief the label had induced more than its evaluative penalty. A residual penalty appeared only for artworks that participants still recognized, and only on the categorical choice between ugly and beautiful, the component through which the labels had shifted beliefs in Phase 2. And the syntheticness belief formed in Phase 2 carried over: given its Phase 1 beauty, an artwork that a participant had judged more human-made was found more beautiful six weeks later, so that the AI label kept a small indirect hold on appreciation through the belief it had created.

What participants explicitly remembered showed the same dissociation between the label and the belief it induced, more starkly. They did not remember which label an artwork had carried: their attributions were at chance and followed a response bias toward the Original label, perhaps encouraged by the reminder that all artworks were genuine. They did remember what they had believed, along the syntheticness dimension at least, beyond what their current impression of the work could explain, and they reconstructed the label from this belief and from their Phase 1 appraisal: artworks they had judged AI-generated, or had found less beautiful, were more often remembered as labelled AI-Generated. In the terms of the source-monitoring framework (Johnson et al., 1993), the content of the belief survived while its source was lost, and the source was then inferred from the content: participants attributed to the label what they had concluded themselves. For the disclosure of AI-generated content, this means that a label may be forgotten within weeks while the beliefs it shaped remain, and that the labels people remember having seen reflect their own conclusions as much as the information they were given. For research, it means that delayed checks of a labelling manipulation measure beliefs as much as memory for the labels.

## An Attitude, Not a Sensitivity

Participants differed reliably in their baseline judgments but much less in their sensitivity to the labels. The effect of the AI label was a moderately reliable individual characteristic, that of the Forgery label only on worth, and individual differences in the label effects on beliefs could not be resolved with sixteen artworks per label; studies of individual differences in the bias will need many more trials, especially for beliefs. Where it could be measured, the anti-fake bias formed a dimension of its own, largely independent of how much participants appreciated the artworks, and was related, as preregistered, to attitudes toward AI and to nothing else: participants with more negative attitudes toward AI, and those who denied that AI-generated art can rival human art, penalized the fake labels more, whereas art expertise, knowledge and use of AI tools, and distrust of the manipulation played no role. The bias follows what people think about AI, not what they know about art, as a belief-driven account predicts.

Contrary to our preregistered predictions, neither interoceptive sensibility nor the vividness of visual imagery was related to the bias. Interoceptive sensibility was related instead to the baseline: participants reporting more interocep-

tive awareness found the artworks more beautiful, pleasant, meaningful and valuable, in line with work linking aesthetic experience to interoception (Cabbai et al., 2023; Washizu et al., 2025), and made more decisive reality judgments on both sides of the scales. In the Choice-Confidence model, this decisiveness is the confidence component of the judgment, how far toward the end of the chosen side the response goes. Work on interoception and metacognition suggests that confidence not only draws on sensory evidence but on bodily states as well (see Allen et al., 2016; Banellis et al., 2025, 2026). Interoception questionnaires may therefore partly measure a general confidence in, or reliance on, one's felt impressions, which would show wherever a judgment is explicitly left to gut feeling, as the reality judgments were. As in interoceptive tasks, this confidence came without accuracy: all artworks were human originals, yet participants reporting more interoceptive awareness were also more convinced when they judged an artwork AI-generated or a copy. Response styles can produce spurious correlations between questionnaires and confidence ratings (Sarna et al., 2026), but this one was specific to the reality judgments (interoceptive sensibility was unrelated to the extremity of ugliness judgments), and the questionnaire data of inattentive participants had been discarded. Interoception may thus shape how strongly people engage with art and how confidently they judge its origin.

## Distinct Dimensions of Aesthetic Experience

Subjective aesthetic dimensions are often highly correlated and collapsed into a single appreciation factor. The present data argue for keeping them apart. Beauty, valence, meaning and worth map onto distinct processes, from sensory evaluation to affect, semantic processing and reward, and each showed a distributional signature of its own: valence symmetric and unimodal; meaningfulness zero-inflated, as if works were first gated as having meaning or none; worth right-skewed and decaying, as in economic valuation; beauty bimodal, combining polarized judgment with the anchoring that analog sliders invite. Response formats contributed to these shapes, but the results support the distinction: the labels lowered meaningfulness most and drove it to outright denial, and only beauty and meaningfulness carried the label effects on beliefs. Modelling each outcome with a family suited to its shape also separated the categorical from the graded components of the judgments, showing that the labels lowered both the probability of finding a work beautiful and its degree of beauty, shifted beliefs through the categorical choice alone, and left their residual effect at follow-up in the categorical component only.

## Limitations

All stimuli were genuine human paintings, which isolates the effect of belief but leaves open how labels interact with the features of actual AI-generated images; the AI label may also have been less plausible for some paintings than for others, although distrust of the manipulation was unrelated to the bias. The Forgery label described copies made to deceive, so its penalty may include a moral disapproval of fraud, and the AI label may likewise carry moral connotations, such as the appropriation of artists' work, which its association with attitudes toward AI does not separate from aesthetic expectations; a label describing an honest reproduction would isolate non-originality more cleanly. The labels did not cross syntheticness and authenticity: AI originals and AI copies were distinguished in the beliefs but not in the labels. The decomposition of the label effects on beliefs rests on assumptions that randomizing the label does not secure: appraisal was measured, not manipulated, and measurement error in the ratings would understate the indirect effects and overstate the direct ones. The pseudo-reveal of Phase 2 left the labels a partially valid cue, by design, so that the Phase 2 label effects cannot be read as a perseverance of beliefs. The follow-up confounded the passage of time with the correction, was completed by about two thirds of the sample, and was the only session in which self-relevance was measured.

## Conclusion

The devaluation of art attributed to AI combines a generic penalty for works that are not authentic originals, shared with human forgeries, with a larger penalty specific to the absence of a human author, which falls most heavily on the meaning a work is granted. This penalty is translated into beliefs about origin and reinforced by them, as viewers take beautiful and personally resonant works to be human-made, labelled or not. It is indifferent to the affective content and personal relevance of the work, and follows people's attitudes toward AI rather than their sensitivity to art or to their own body. Once the label is withdrawn and corrected, its effect on judgment fades, but the beliefs it induced outlast the memory of the label, and are later remembered as what the label said.

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